

AI-Assisted Communication, Trust, and Coordination in Human Partnerships: Evidence from a Trust Game Experiment ^{*}

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Abstract

We experimentally study whether AI-assisted communication facilitates trust between humans in a two-player trust game with communication before play, in which a trustor decides whether to enter a risky interaction and a trustee decides whether to reciprocate. In a 2 x 2 online experiment with 240 Prolific participants from the United States and the United Kingdom, with 30 pairs per condition, we vary whether the trustee receives GPT-3.5-turbo-0613 assistance in composing a message and whether the trustor receives AI assistance in interpreting it. We find limited evidence that AI access, whether provided to the trustor, the trustee, or both, directly changes trustors' entry decisions, trustees' reciprocation decisions, or trustors' beliefs about whether trustees will reciprocate. However, trustee AI assistance changes communication: trustees with AI access are more likely to send promises to reciprocate or requests to enter, while there is suggestive evidence that AI assisted promises are honored less often than self-authored ones. Even so, the frequency of cooperative outcomes, in which the trustor enters and the trustee reciprocates, increases directionally when trustees have access to AI. Taken together, these results point to a possible pair-level coordination effect even when average individual choices and beliefs do not shift strongly. Because the experiment uses 30 pairs per condition and one specific LLM version, the findings should be interpreted with appropriate caution.

Key Words: Trust, AI, ChatGPT, Experiment

JEL Codes: C91, C72.

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1 Introduction

Trust is a cornerstone in various socioeconomic activities, including partnership formations and financial transactions, where it transcends mere contractual obligations. It is vital in relationships ranging from personal bonds, like those between spouses, to professional associations, such as the lawyer-client dynamic, procurement agencies and contracted firms, and collaborations between researchers and participants in scientific studies (Charness and Dufwenberg, 2006). Extensive research demonstrates its significance in several financial dealings: it influences stock market participation (Guiso et al., 2004, 2008), affects consumer credit (Brown et al., 2019), determines the use of investment advisers (Gurun et al., 2017), and helps foster healthy lending relationships between lenders and borrowers in the credit markets (Fisman et al., 2017, Hyndman et al., 2024).

Building trust relies on successful communication among involved parties. It is particularly dependent on the ability of the trustee to convince the trustor to place their trust in them. This becomes challenging when the communication is non-binding and the interactions are not regulated by formal agreements. Existing experimental studies show that pre-play communication, especially promises, can facilitate trust building in partnership formation (Charness and Dufwenberg, 2006).

In the era of artificial intelligence, large language models (LLMs) like the Generative Pre-trained Transformer (GPT) have become integral to human communication.¹ These models assist in various tasks such as drafting emails, generating financial reports, improving academic papers, and editing other written materials.² When such assistance is used in communication between people, AI becomes part of the communication process and may shape not only the content of a message but also how its authorship, credibility, responsibility, and commitment are perceived. This raises a pertinent question: what role can AI play in

¹GPT is created by OpenAI, significantly influencing the field of natural language processing (OpenAI, 2022, 2023b). Brown et al. (2020) show that ChatGPT can produce texts with such remarkable accuracy and fluency that it closely resembles human writing, making it challenging for human evaluators to differentiate between text generated by GPT and that authored by humans.

²In addition, LLMs have shown remarkable capabilities across diverse areas. They are capable of creating computer code, as demonstrated by Chen et al. (2021), and solving university-level mathematics problems (Drori et al., 2022).

the trust-building process between a trustor and a trustee? The answer is not obvious, yet a clearer understanding of this question is essential for practical applications. For instance, in a bank loan application, a borrower might use AI to compose their application, while the lender might use AI for interpreting the application. The advantages of AI in this context are clear: it saves time for both parties, enhances the clarity and substance of the borrower’s application, and helps the lender quickly understand the key points. However, AI also has potential drawbacks. It could diminish the authenticity of the borrower’s application, leading the lender to question its veracity. Moreover, if the borrower heavily relies on AI, they might feel less committed to the content produced by the AI, posing a risk to the lender.

This study aims to explore the impact of AI-mediated communication on trust-building in a controlled experimental setup. We use a two-player binary-choice sequential trust game based on [Charness and Dufwenberg \(2006\)](#). Player A, the trustor, first decides whether to enter a risky interaction. Player B, the trustee, then decides whether to reciprocate. Before the game begins, Player B may send a free-form message to Player A. We implement a 2×2 design. In one dimension, Player B either has or does not have access to AI assistance when composing the message. In the other dimension, Player A either has or does not have access to AI assistance when interpreting Player B’s message.³ This design allows us to study whether AI assistance affects the production of messages, trustors’ beliefs about trustees’ behavior, trustees’ promise-keeping, and the cooperative outcome in which Player A enters and Player B reciprocates.

We find that AI assistance does not significantly change the overall frequency with which trustors enter the interaction or trustees reciprocate. However, AI assistance changes communication. When trustees have access to AI, they are somewhat more likely to send messages that contain promises to reciprocate or requests for the trustor to enter. At the same time, promises that originate from AI suggestions are honored less often in the sample than promises authored by trustees themselves, although this evidence is exploratory because the relevant subsample is small. We also observe tentative evidence that trustee-side AI

³We employ the GPT-3.5-turbo-0613 model ([OpenAI, 2023a](#)) in the experiment. We carefully design prompts to ensure AI fully comprehends the trust game and understands its role as an assistant for a specific player in the game during the communication phase. See [OpenAI \(2023c\)](#) for guidance on prompt design.

assistance increases the frequency of cooperative outcomes, consistent with the possibility that AI assistance may reduce miscoordination by helping trustors who are willing to respond to credible communication and trustees who are willing to act trustworthily align on the cooperative outcome.

The paper makes three contributions. First, it contributes conceptually to research on trust and AI-mediated communication by developing a framework that links AI assistance in message production and interpretation to trustor beliefs, trustee trustworthiness, and coordination between humans, rather than trust in AI itself. Second, it provides experimental evidence on how AI assistance affects voluntary commitments and cooperative interaction, including promise-making, promise-keeping, beliefs about trustworthiness, and pair-level coordination on the cooperative outcome. Third, it adapts a classic trust-game paradigm to an HCI-relevant setting in which AI is neither the decision-maker nor the object of trust, but a communication aid embedded in human-to-human interaction

The paper is organized as follows. Section 2 provides the experimental design and proposes the main hypotheses. Section 3 analyzes the experimental results and test the hypotheses. Section 4 concludes.

Literature Review

Trust and trustworthiness are related but distinct concepts, and different literatures have converged on complementary ways of defining them.

In organizational research, [Mayer et al. \(1995\)](#) define trust as one party’s willingness to be vulnerable to another party’s actions, based on the expectation that the other party will perform a particular action important to the truster, regardless of the ability to monitor or control that party. [Rousseau et al. \(1998\)](#) extend this across disciplines, describing trust as a psychological state comprising the intention to accept vulnerability based on positive expectations of another’s intentions or behavior. These foundational definitions establish the two elements that recur across the literatures below: a willingness to accept vulnerability, and an expectation about the other party’s future conduct.

Information systems research applies this same idea to online vendors, platforms, and digital environments. [McKnight and Chervany \(2001\)](#) and [McKnight et al. \(2002\)](#) decompose trust into trusting beliefs – perceptions that the other party is competent, benevolent, of high integrity, and predictable – and trusting intentions, the willingness to actually depend on that party in a risky transaction. [Gefen et al. \(2003\)](#) and [Pavlou \(2003\)](#) apply this belief-intention structure to online consumers, showing that trusting beliefs about a vendor predict the intention to transact with it, even without prior interaction.

A related literature in HCI and human-automation research studies trust as a person’s reliance on a technological agent, such as an automated system or a robot, rather than on another human. [Lee and See \(2004\)](#) argue that trust in automation depends on the system’s perceived performance, process, and purpose, and that appropriate trust requires calibrating reliance to the system’s actual reliability. [Hoff and Bashir \(2015\)](#) synthesize this literature and show that trust in automation is shaped jointly by stable dispositional factors, the situation at hand, and the system’s demonstrated performance over time. [Hancock et al. \(2011\)](#) find, in a meta-analysis of human-robot interaction studies, that a robot’s performance characteristics are the strongest predictor of a person’s trust in it, more so than the robot’s appearance or the person’s own traits.

Operations and behavioral operations research draw a related distinction in supply-chain and service settings, where formal contracts are typically incomplete. Here, trust refers to one party’s reliance on a counterpart despite that incompleteness, and trustworthiness refers to the counterpart’s tendency to justify that reliance—for example, by sharing information truthfully, honoring informal commitments, or sustaining high-quality performance even when opportunistic behavior would be more profitable ([Özer et al., 2011, 2014, Beer et al., 2018](#)).

Experimental trust games in economics provide a widely used paradigm for studying trust and trustworthiness in settings where cooperation is mutually beneficial but not contractually enforceable. In this literature, cooperation refers to the joint outcome in which the trustor enters a risky interaction and the trustee reciprocates by taking a costly action that benefits the trustor. The trustor’s belief about whether the trustee will reciprocate captures trust, while her decision to enter reflects her willingness to act on that belief despite the risk of opportunistic

behavior. The trustee’s decision captures trustworthiness: she must decide whether to reciprocate the trustor rather than exploiting the trustor’s vulnerability. Coordination refers to the alignment of the trustor’s and trustee’s actions on the cooperative outcome. Pre-play communication is especially important in such environments because it allows the trustee to communicate intentions before the trustor makes the entry decision. The trust-game paradigm has been extensively studied, as has pre-play communication in other strategic settings. [Charness and Dufwenberg \(2006\)](#) brought these two established experimental approaches together by introducing non-binding pre-play communication into a trust game, establishing a paradigm that has since become widely used. They find that trustee messages substantially increase cooperation, especially when they contain promises to reciprocate. Their interpretation is guilt aversion: a promise raises the trustor’s expectation of cooperation, making promise-breaking psychologically costly for the trustee. [Vanberg \(2008\)](#) instead emphasizes promise-keeping itself, whereby individuals prefer to keep their word even absent belief-based guilt. [Charness and Dufwenberg \(2010\)](#) show that free-form promises are more effective than bare, pre-scripted ones, while [Di Bartolomeo et al. \(2019\)](#) further distinguish promise-keeping from belief-dependent guilt. [Cominelli et al. \(2021\)](#) extend the paradigm to human–robot interaction, but it has not yet been applied to AI as a communication aid between human partners. This literature highlights three concepts central to our study: communication about intentions; promise credibility to the trustor, or how informative the promise is about future reciprocation; and the promise’s behavioral force for the trustee, or how strongly making it increases the likelihood that it will be honored.

AI assistance may alter these trust-game mechanisms because it changes how pre-play messages are produced, attributed, and interpreted. In AI-mediated communication, a message can be written by the trustee herself, suggested by AI, or jointly produced by both. This may improve clarity and persuasiveness by helping trustees formulate intentions more explicitly, making it easier for trustors and trustees to coordinate on the cooperative outcome. At the same time, research on AI-mediated communication suggests that AI involvement can change the social meaning of a message. [Hancock et al. \(2020\)](#) argue that when a machine intervenes between two communicating parties—whether by modifying, augmenting, or generating a

message—it becomes difficult to say whether the human or the system is the true author, which complicates who is held responsible for what gets said. [Sundar \(2020\)](#) frames this as a shift in perceived “machine agency”: as people come to see AI as an independent source of a message rather than a passive tool, they attribute less of the message’s intent to the human sender. Consistent with this, [Jakesch et al. \(2019\)](#) find experimentally that simply believing a profile description was written by AI—regardless of who actually wrote it—lowers how trustworthy the writer is judged to be. [Hohenstein and Jung \(2020\)](#) describe a related pattern in which AI mediation of a message creates a “moral crumple zone”: responsibility for the message’s content becomes diffused between the human and the system in a way that lets the human sender partially disclaim ownership of it. Together, these findings suggest that AI assistance can blur the division of agency between the human and the system and shift how authorship and responsibility for a message are attributed.

This shift in perceived authorship and responsibility may, in turn, weaken the trustee’s own commitment to the message: a promise that is not entirely the trustee’s own words may carry a weaker personal stake, since a sense of obligation to keep a promise is bound up with viewing it as one’s own, echoing broader work on how responsibility for an outcome is diffused when more than one party could have acted ([Darley and Latané, 1968](#)), how a ready-made excuse can create moral “wiggle room” that people are quick to exploit ([Dana et al., 2007](#)), and how people invest less of their self-image in commitments they did not fully originate ([Bénabou and Tirole, 2011](#)). This creates a theoretically ambiguous effect. AI-assisted messages may strengthen trust by making intentions clearer and facilitating coordination, but they may also weaken trust by reducing the perceived credibility and personal force of voluntary promises.

Our paper is closest to work in which AI assists human participants rather than replacing them as autonomous decision-makers.⁴ In [Harris et al. \(2023\)](#), AI provides information that

⁴A growing literature examines AI as a decision-maker in human activities. One strand uses surveys, laboratory-style experiments, and strategic games to examine whether AI systems reproduce human-like behavior in individual decisions and interactive settings ([Aher et al., 2023](#), [Argyle et al., 2022](#), [Bybee, 2023](#), [Brand et al., 2023](#), [Brookins and DeBacker, 2023](#), [Chen et al., 2023](#), [Fan et al., 2023](#), [Guo, 2023](#), [Hagendorff, 2023](#), [Horton, 2023](#), [Kosinski, 2023](#), [Lorè and Heydari, 2023](#), [Ma et al., 2023](#), [Phelps and Russell, 2023](#), [Engel et al., 2024](#), [Leng and Yuan, 2023](#), [Strachan et al., 2024](#)). A second strand studies human interaction with machines or AI agents, including whether people respond differently to algorithmic counterparts than to human counterparts, whether AI agents outperform humans, and whether human-AI interaction creates

helps one person decide how to communicate with another person whose decision they hope to influence. [Bai et al. \(2023\)](#) studies whether people rely on AI advice when making sequential decisions in a two-person strategic interaction. [Serra-Garcia and Gneezy \(2023\)](#) show that algorithmic tools can help people detect deception. These studies show that AI can shape information acquisition, strategic reasoning, and deception detection.

Our setting asks a different question, one that remains underexplored in the existing literature: how AI-assisted communication affects human-to-human trust formation when the relevant message is a voluntary, non-binding commitment. This question is distinct from three related literatures in which the AI itself is the object or source of reliance. Research on trust in automation examines whether people rely on an automated system and how that reliance responds to its reliability ([Glikson and Woolley, 2020](#)). Research on algorithm aversion studies reluctance to rely on algorithmic judgment, especially after observing errors ([Dietvorst et al., 2015](#)). Research on AI advice-taking examines whether people follow AI-generated recommendations for their own decisions ([Logg et al., 2019](#)). In our setting, by contrast, trust remains between humans, the trust-relevant choices remain human choices, and AI assists communication before those choices are made. By changing message content and interpretation, AI may affect not only trust but also coordination on the mutually beneficial outcome.

In summary, [Figure 1](#) presents the conceptual framework linking AI assistance to message production and interpretation, trustor beliefs, trustee behavior, and coordination. The next section describes how these relationships are examined in our experiment.

2 The Experiment

2.1 The Trust Game Environment and Operational Measures

The objective of this study is to explore the potential of AI assistants in fostering trust dynamics between human participants. To accomplish this goal, we conducted an online

agency or accountability problems ([Bauer et al., 2023](#), [Cohn et al., 2022](#), [de Mello et al., 2016](#), [Phelps and Russell, 2023](#), [LaMothe and Bobek, 2020](#), [Laudenbach and Siegel, 2024](#), [Schniter, 2024](#), [Dvorak et al., 2024](#)).

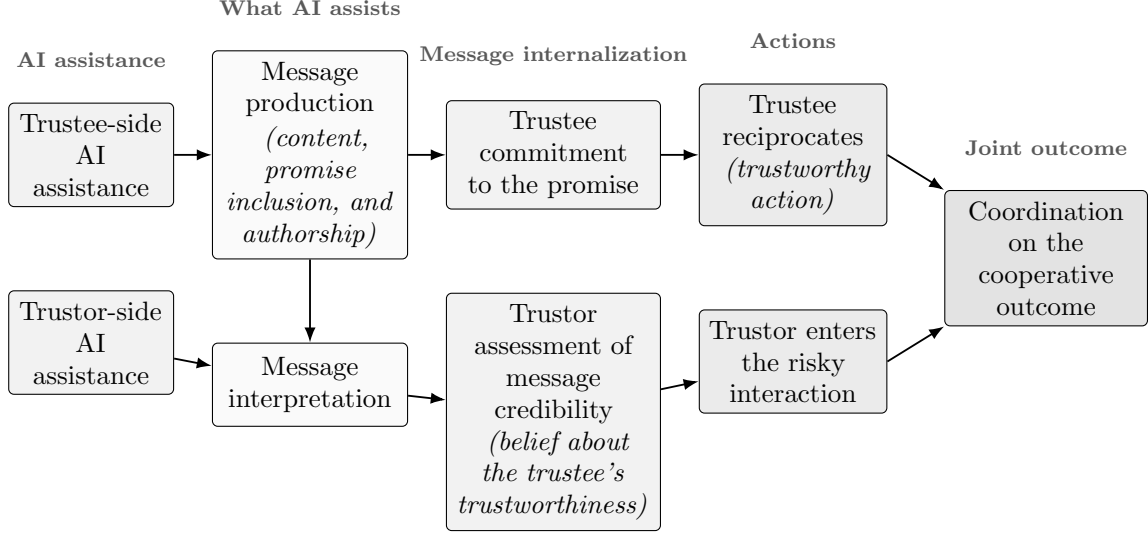


Figure 1: Conceptual model of AI-assisted communication, trust, and coordination.

experiment based on the classic two-player trust game introduced by [Charness and Dufwenberg \(2006\)](#), by allowing the second player to communicate with the first player via a pre-game message. The theoretical motivation for using this paradigm is discussed in the literature review; here we describe the game environment and the operational measures used in the experiment.

This trust game can be viewed as a principal–agent partnership over a risky project. Player A (the trustor) is the principal who decides whether to form the partnership by choosing ‘In’ or to walk away by choosing ‘Out’. If she chooses ‘Out’, no partnership is formed and both players take their outside option of 7. If she chooses ‘In’, the partnership is formed: Player B (the trustee) acts as the agent who receives a wage of 14 and then chooses an effort level correlated with the outcome of the project. Low effort (‘Don’t Roll’) is costless to player B but yields no return so Player A receives 0 while Player B ends up with 14; should player B choose high effort (‘Roll’), it costs her 4 and generates a stochastic return for Player A—12 with probability $5/6$ and 0 with probability $1/6$ —leaving player B with 10 for sure. The resulting sequential interaction is depicted using a game tree in [Figure 2](#), with outcomes shown in the order of (π_A, π_B) .

Notice that the ‘Roll’ option leads to an expected payoff of 10 for Player A. Hence, compared to the situation led by Player A choosing ‘Out’, (‘In’, ‘Roll’) improves both players’ expected

payoffs. However, if Player B is self-interested, she would choose ‘Don’t Roll’ to get the higher payoff of 14. By anticipating this, Player A would choose ‘Out’ to protect herself.

We can regard ‘Out’ as the no trust situation, while (‘In’, ‘Roll’) represents the case that trust fosters cooperation and yields a higher expected payoff for everyone. However, cooperation is hard to sustain because of selfish temptation.

If Player B is not driven purely by monetary outcome and is willing to sacrifice some of her own reward to increase Player A’s payoff, then cooperation may be achievable. A main obstacle remains, however: Player A is uncertain about how likely Player B is to choose ‘Roll’, which makes it difficult for Player A to place trust in Player B. To formalize this idea, we introduce the following operational definitions.

Definition 1.

- (1) Player B’s *trustworthiness* is measured by her likelihood of choosing ‘Roll’.
- (2) Player A’s *trust* toward Player B is measured by Player A’s belief of Player B’s likelihood of choosing ‘Roll’.

In our experimental setting, trust is not measured by a self-reported attitude but is operationalized in a decision-theoretic way through beliefs and subsequent choices as given in Definition 1. The reason is that “trust” is inherently about expectations regarding another person’s behavior in a strategic situation. A direct survey question asking participants how much they trust their counterpart would be cheap talk: it is not incentive compatible, is sensitive to wording and demand effects, and need not map into behavior in the game. By contrast, eliciting Player A’s belief about Player B’s likelihood of choosing ‘Roll’ captures the object that is central to the trust decision. Player A’s willingness to choose ‘In’ is then a behavioral manifestation of that belief, but it is confounded by her risk attitude.⁵ Using beliefs therefore provides a cleaner, more targeted measure of trust than relying on the action ‘In’ alone, while Player B’s propensity to choose ‘Roll’ provides the corresponding behavioral

⁵If Player A is extremely risk-averse, she may choose ‘Out’ even if she believes Player B is very likely to choose ‘Roll’. Conversely, if Player A is extremely risk-seeking, she may choose ‘In’ even if she expects Player B to choose ‘Roll’ with low probability. We explore this mechanism more deeply in Section 3.3.

measure of trustworthiness.

This operationalization necessarily narrows trust to Player A’s belief about one specific choice by Player B. It therefore does not capture every facet of trust emphasized in the broader organizational, information systems, and HCI literatures, including trust grounded in prior interactions, relational history, or vulnerability. This narrower operationalization is appropriate for our one-shot experimental setting because the participants have no prior relationship, no shared history, and no future interaction. The relevant uncertainty is therefore whether Player B will reciprocate in the current interaction. This setting maps naturally onto AI-assisted communication on gig platforms such as Upwork, where a client with little or no prior relationship with a worker must decide whether to proceed based partly on a message promising a particular level of quality or a deadline. In such settings, trust centers on a positive expectation about a specific future action, and the decision to proceed exposes the trustor to potential loss. Accordingly, our measure isolates the belief component of broader conceptualizations of trust, while Player A’s decision to enter captures her willingness to accept vulnerability based on that belief.

Integral to the context of our study, trust and trustworthiness are defined between human participants, not between a human and an AI. The presence of AI may affect Player B’s behavior and Player A’s belief and behavior, but it does not alter the object of trust: Player A is asked to form expectations about Player B, not about the AI assistant. In this sense, AI enters the experiment as a communication aid that may shape human beliefs and behavior, rather than as the party being trusted.

An important design feature of the [Charness and Dufwenberg \(2006\)](#) is a pre-play communication stage. Before Player A makes her decision, Player B is allowed to send a single free-form message to Player A, with no further message exchange. This message gives Player B an opportunity to communicate her intentions and gives Player A information that may shape her belief about whether Player B will choose ‘Roll’. Because the message is voluntary and non-binding, this stage is especially useful for studying whether AI assistance affects the trust-building process through communication. In our experiment, we introduce AI assistance at this stage: Player B may receive AI assistance when composing the message, and Player A

may receive AI assistance when interpreting it.

2.2 Experimental Design

2.2.1 Treatment Design

To investigate the impact of AI assistance on players' payoffs in the trust game, we implemented a 2×2 treatment design, consisting of four distinct treatments. Each treatment explored various combinations of AI presence and absence, shedding light on the role of AI in participants' decision-making processes and outcomes.

- **Benchmark:** Neither player has AI. Player B can choose to send a single message to Player A or refrain from doing so.
- **OnlyA:** Player A has AI to interpret Player B's (potential) message and receive advice on subsequent actions.
- **OnlyB:** Player B has AI to assist in crafting a message to Player A. If Player B opts to send a message, they must first interact with the AI.
- **Both:** Both players have access to AI assistance, following the functionalities described above.

In all treatments, player B has the option to send a message to player A prior to the game's start. This message is completely voluntary and player B is free to decide its content. When applicable, player B's interaction with AI precedes any communication with Player A. While Player A does not directly see whether player B actually uses AI assistance, it is crucial to note that all participants are aware of the presence and function of AI throughout the experiment. A depiction of player B's interface with AI assistance is given in Figure 2.

In sum, our experiment design involves a one-shot, anonymous trust game with optional pre-play messaging and no enforcement or reputation. AI assistance is provided only for composing messages (and is common knowledge), so our results speak to AI-mediated communication in short, low-information interactions rather than long-term relationships or settings with formal contracts.

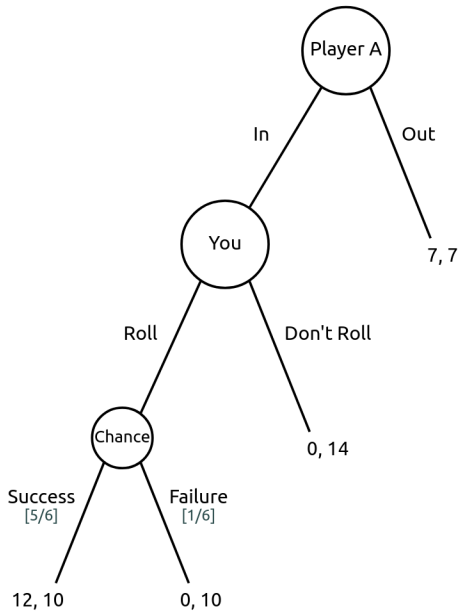
Information:

This is a 2-player game. **You are Player B.** Player A moves first. Player A can choose Out or In. If they choose Out then both players get a payoff of 7. If they choose In, it is player B's turn to move. Player B can choose either Roll or Don't Roll. If player B chooses Don't Roll, player A will get a payoff of 0 and player B will get a payoff of 14. If player B chooses Roll, they will receive a guaranteed payoff of 10 while player A has a 5/6 probability of receiving 12 and 1/6 probability of receiving 0.

Instructions:

You can send one message to Player A. Before sending the message, an AI assistant will give you feedback on your message. You may choose to ask the AI for help or send to the AI the message you intend to send to Player A.

Choose not to send a message to Player A

**Draft a message to the other player:**

Draft a message to send to player A or ask the AI for advice in composing the

Send your message to AI

Send your message to Player A

AI Assistant:**AI Suggested Message (you can edit this):**

The AI suggested message will appear here.

Send AI assisted message to Player A

Figure 2: Screenshot of player B's screen when they are able to use AI. Player A sees a similar tree.

2.2.2 Additional Experimental Details

To maximize data collection and ensure simultaneous decision-making by both players, we employed the strategy method (Selten, 1967). In particular, Player B specifies contingent actions in advance, i.e., how she would respond if Player A chooses 'In', regardless of whether that node is ultimately reached. This allows us to observe Player B's intended behavior even in instances where Player A chooses 'Out', and it avoids waiting for Player A's move before eliciting Player B's decision, thereby keeping choices effectively simultaneous across participants.

Following each player’s decision, Player A is asked to report her beliefs about Player B’s choice, while Player B is asked to report her beliefs about Player A’s expectations. Belief elicitation from Player A provides a measure of her trust, while belief elicitation from Player B allows us to examine how her belief influences her decision.

Subsequently, both participants are presented with a Holt-Laury quiz ([Holt and Laury, 2002](#)) to assess their risk preferences. Player A’s risk preference is particularly important because it affects a Player A’s decision of choosing ‘In’. For example, an extremely risk-averse Player A may choose ‘Out’ even if she highly trusts that Player B would choose ‘Roll’.

Additionally, a demographic survey was administered to gather information on participants’ perceptions and prior experience with AI technology. This may allow us to assess baseline heterogeneity in AI familiarity and attitudes that may shape how participants use AI assistance, how credible they perceive AI-assisted messages to be, and how they interpret and respond to their counterpart’s communication.

2.2.3 Prompt Design

Creating an AI assistant using a natural language processor like ChatGPT involves crafting a prompt that guides the assistant in generating responses. However, designing a prompt that effectively handles a wide range of inputs is more of an art than a science. Our experimentation with ChatGPT (specifically gpt-3.5-turbo-0613) revealed certain challenges and considerations that influenced the development of robust prompts for our study.

We observed that ChatGPT exhibited difficulties in handling sequential logic and tended to perform more reliably with shorter prompts. Moreover, we noted instances where the assistant suggested creative solutions, such as proposing the signing of a contract to establish a binding promise, which were beyond the scope of our experimental setting.

In light of these observations, we refined our prompts by incorporating the following principles:

1. We presented the trust game in its normal form, as the sequential form did not strategically differ from its normal counterpart given the strategy method employed in our experiment.

2. We omitted the description of probabilistic outcomes resulting from ('In', 'Roll') in the game for Player B's AI only and instead provided ChatGPT with the expected outcomes.⁶
3. We explicitly instructed ChatGPT not to propose side deals or disclose players' personal information.
4. We refrained from including higher-order beliefs for the AI assistants. Specifically, Player B's AI did not possess knowledge of the existence of Player A's AI.

At their core, the prompts instructed Player B's assistant to help the trustee craft a persuasive message to Player A with the goal of maximizing Player B's payoff, while Player A's assistant was instructed to interpret Player B's message and advise Player A which action to choose with the goal of maximizing Player A's payoff.

In practice, Player A's assistant often adopted a cautious tone, and strategically inconsistent messages could confuse its interpretation; representative examples are provided in Appendix A.2.3.

The full prompts used in our study can be found in Appendix A.3.

2.3 Experimental Procedure

A total of 240 subjects, 30 pairs per treatment, were recruited from to participate in this experiment. On average, subjects spent 15 minutes in the experiment and were paid a \$5 show up fee and earned an additional \$8.36 during the experiment⁷. The experiment was programmed using oTree (Chen et al., 2016) and ChatGPT version gpt-3.5-turbo-0613. It was implemented on Prolific.

During the online experiment, subjects were continuously recruited and dynamically assigned roles and partners to play the game with to minimize subject wait time. Subjects read instructions then were assigned their role. They then took a quiz to ensure they understood

⁶We still keep the probabilistic outcomes resulting from ('In', 'Roll') in the game for Player A's AI because we want the AI is able to remind Player A to take risks into consideration.

⁷The equivalent of \$33.44/hour, on average. Prolific maintains an \$8/hour minimum that researchers must pay to subjects.

the payoff structure before getting paired with a partner and playing the game. An example of the experimental instruction can be found in Appendix A.8.

2.4 Hypotheses

In this section, we lay out the main hypotheses that we tested in the experiment, organized around the relationships summarized in Figure 1. Each hypothesis states a theoretical prediction, which we then operationalize in terms of the specific choice or belief it corresponds to in the game, motivate with the theoretical reasoning behind it, and follow with a discussion of what a supported or unsupported result would imply.

Hypothesis 1: *Trustee-side AI assistance increases the trustor’s willingness to enter the risky interaction.*

Operationally, it means that when player B has access to an AI, player A will play ‘In’ more frequently. We also examine whether Player B’s access to AI increases Player A’s elicited belief that Player B will choose ‘Roll’, which measures Player A’s trust toward Player B. The hypothesis is motivated by our theoretical construct that AI assistance can improve the effectiveness of the pre-play message: if AI primarily helps Player B formulate clearer and more informative communication about intended behavior, it can reduce Player A’s strategic uncertainty and thereby increase the attractiveness of choosing ‘In’. More specifically, clearer communication may increase Player A’s belief that Player B will reciprocate, which in turn increases Player A’s willingness to enter the risky interaction. This is in line with existing evidence that AI can generate persuasive messages that shift human attitudes and behavior (Bai et al., 2025, Salvi et al., 2025). At the same time, this force may be offset if AI use makes the message appear less authentic or less personally binding, which could increase Player A’s suspicion. Thus, the predicted increase in ‘In’ rests on the premise that the communication-improvement effect dominates any trust-eroding effect from perceived AI involvement.

Support for Hypothesis 1 would suggest that AI tools used by the party seeking trust

(the party trying to “get the deal”) can increase partnership formation by improving how intentions and commitments are communicated. In settings like hiring, freelancing or lending, AI assistance could raise the rate at which counterparties are willing to move forward (sign, invest, transfer funds) because the communication feels clearer or more complete. If not supported, it points to an important limit: AI-mediated messages may trigger skepticism that offsets any gains from clarity. Because we do not directly measure Player A’s perception of message clarity or whether the message reflects Player B’s own intentions, support for Hypothesis 1 would be consistent with the proposed pathway from clearer communication to stronger beliefs about Player B’s trustworthiness, but would not conclusively establish that pathway.

Hypothesis 2: *Trustor-side AI assistance decreases the trustor’s willingness to enter the risky interaction.*

In practice, this predicts player A playing ‘In’ less frequently when they have an AI assistant. Player A’s AI assistant is instructed to help its user maximize her payoff. If the assistant applies the classic backward-induction logic in game theory and assumes Player B is self-interested, it will predict ‘Don’t Roll’ and therefore recommend ‘Out’ as Player A’s best response, leading to fewer ‘In’ decisions. More specifically, the AI may lead Player A to interpret Player B’s message more cautiously by emphasizing Player B’s monetary incentive to choose ‘Don’t Roll’.

If Hypothesis 2 is supported, it indicates that AI can push decision makers toward more conservative choices by highlighting downside risk and strategic vulnerability. More broadly, it would suggest that AI may reduce cooperation not by changing preferences, but by shifting attention and beliefs about the other side’s incentives. In investment settings, this would translate into fewer deals, as investors may default to safer outside options, especially when enforcement is weak or opportunism is plausible. Player A’s interpretation of the AI’s advice is not something observed in our data. Therefore, support for Hypothesis 2 would be consistent with greater attention to strategic risk but would not conclusively establish that pathway.

Hypothesis 3: *Trustee-side AI assistance increases the trustee’s use of explicit promises in pre-play communication.*

This translates into the prediction that when player B has an AI assistant, they will be more likely to promise to choose ‘Roll’. Player B’s AI assistant is instructed to help its user to maximize their payoff, which means the AI has an incentive to help Player B convince Player A to play ‘In’. As emphasized in the pre-play communication literature, an explicit promise communicates Player B’s intended behavior and can shape Player A’s expectation of whether Player B will reciprocate. Accordingly, in instances where Player B does not initially make a promise, her AI assistant may suggest adding one. This premise is consistent with evidence that AI can generate usable drafts of interpersonal messages based on the situations (Karinshak et al., 2023, Meng et al., 2025).

Support for Hypothesis 3 would indicate that AI assistance changes behavior at the communication stage by making explicit commitments more likely, which is precisely the channel emphasized in the pre-play communication literature. In particular, it implies that AI can increase the supply of messages intended to build trust even when users would not spontaneously produce them. It would not, however, establish that these additional promises are more credible or that the trustees who send them are more trustworthy. Whether AI-originated promises are subsequently honored is addressed in Hypothesis 4.

Hypothesis 4: *AI-originated promises have weaker behavioral force than self-authored promises.*

In testable terms, when player B’s message to player A contains a promise which originated from the AI, player B is less likely to honor the promise. One possible explanation is that an AI-originated promise creates a weaker sense of personal ownership and commitment than a self-authored promise. Player B may therefore perceive a lower psychological cost of breaking it and be less likely to honor it.⁸

⁸This interpretation is consistent with work on diffusion of responsibility, moral “wobble room,” and self-image investment (Darley and Latané, 1968, Dana et al., 2007, Bénabou and Tirole, 2011).

If Hypothesis 4 holds, it raises a cautionary implication for AI-assisted contracting and financial transactions. Commitments may become cheaper to produce than to keep, which could increase the incidence of overpromising. This could lead to more misunderstandings and disputes, and it may increase the need for stronger verification, authentication, or compliance mechanisms when AI drafting is prevalent. In lieu of an exact measure for Player B’s sense of personal obligation to keep the promise or the psychological cost of breaking it, support for Hypothesis 4 would be consistent with this explanation but would not conclusively identify the underlying mechanism.

Hypothesis 5: *AI assistance improves coordination between the trustor and the trustee on cooperation.*

Concretely, this predicts that the cooperative outcome in which Player A chooses ‘In’ and Player B chooses ‘Roll’ will occur more frequently when Player A or Player B has access to AI assistance.

Unlike the preceding hypotheses, which concern the behavior of one player, Hypothesis 5 concerns the joint outcome produced by the choices of both players. We expect AI to facilitate this outcome by improving coordination. When AI assists communication, it can help convey intentions more clearly; when it assists interpretation, it can help Player A interpret Player B’s message. In either case, AI may reduce miscoordination by making it easier for the players to align their choices on the efficient outcome, effectively serving as a coordination device. This interpretation does not require AI to increase the unconditional frequency with which either player chooses the cooperative action; instead, within the trustor–trustee pairs formed by random assignment, AI may increase the frequency with which each side’s choice ends up aligned with the other’s, producing more cooperative outcomes without necessarily changing either player’s average behavior. If cooperation instead does not increase when AI is present, this would suggest that the potential coordination benefit is offset by skepticism toward AI-assisted communication or greater caution in responding to it. Because we observe the joint cooperative outcome but do not explicitly capture whether AI improves message clarity or changes Player A’s interpretation of Player B’s message, support for Hypothesis 5

would indicate improved coordination but would not identify which specific channel produced it.

2.5 Statistical tests and power

For the purposes of this paper, we primarily use two-sample tests of equality of means to compare outcomes across two independent groups. For binary outcomes (choices), the sample mean is a proportion, and we test for equality of means using the standard two-sample test for equality of proportions (equivalently, a Pearson χ^2 test on the associated 2×2 contingency table). For bounded numeric outcomes in $[0, 1]$ (beliefs), we test for equality of means using Welch’s two-sample t -test, which allows group variances to differ. All tests are two-sided and conducted at the 5% significance level.

For each outcome, we report the estimated difference in means (or difference in proportions), the corresponding test statistic (z for binary outcomes; t with Welch degrees of freedom for bounded numeric outcomes), and the associated two-sided p -value.

Ex ante power and minimum detectable effects. We evaluate detectability ex ante under 80% power and a two-sided 5% test. Under conservative variance assumptions, the implied minimum detectable effect (MDE) for our primary two-group comparisons is approximately 0.25 (i.e., about a 25% increase in the probability of choosing a given action, or a 0.25 increase in the mean belief) for two balanced groups with 60 observations per group. For two unbalanced groups (e.g., 30 observations in one group and 90 in the other), the MDE is modestly larger; we therefore treat 0.25 as a simple conservative benchmark for the magnitude of effects that the study is powered to detect.

3 Results

3.1 Treatment Effects

To investigate the influence of AI assistance on outcomes, Figure 3 presents the decisions made by each player when (i) neither player receives AI assistance (baseline), (ii) player A

receives AI assistance but not player B, (iii) player B receives AI assistance but not player A, and (iv) both players receive AI assistance.

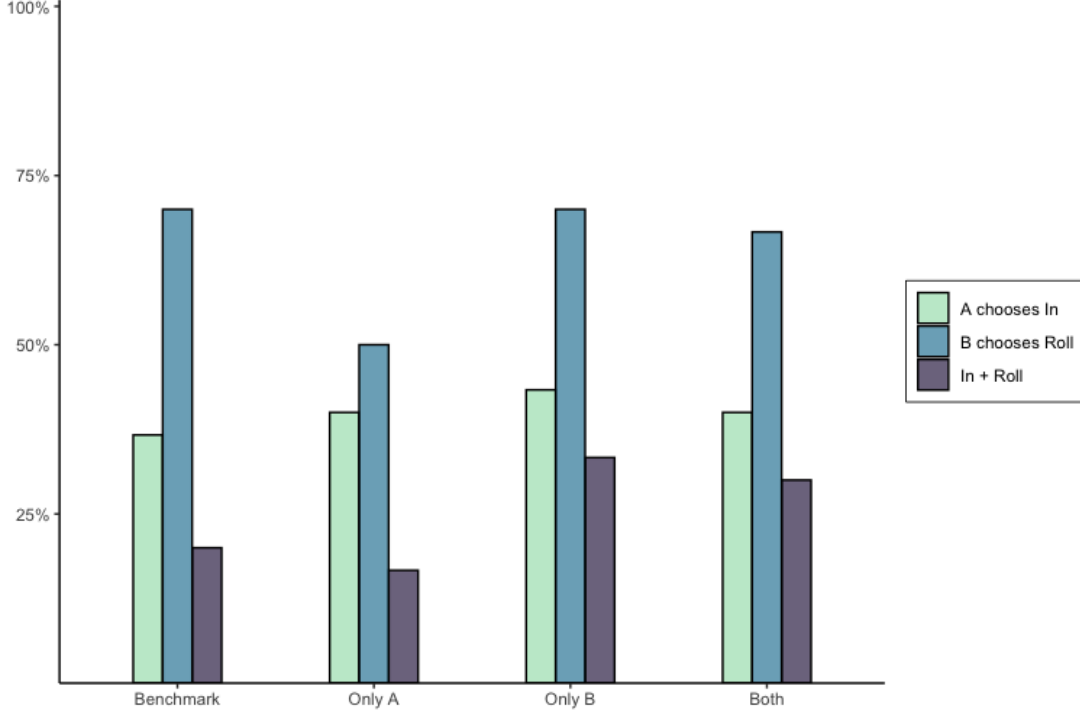


Figure 3: The percentage of observations in each treatment where player A chooses ‘In’, player B chooses ‘Roll’, or both.

In the benchmark treatment, player A chooses ‘In’ 36.7% of the time. This rate increases to 43.3% once player B has the option to use AI. Conversely, in both treatments where player A receives input from AI, 40% of trustors choose ‘In’ whether or not player B has AI. These results point to a negligible impact of AI’s presence on player A’s choices.

We test **Hypothesis 1** by pooling treatments based on player B’s access to AI.⁹ We observe that player A chooses ‘In’ more frequently when player B has AI (41.67%) than without AI (38.33%). However, we fail to reject the null (effect size = 3.34%, $p = 0.709$; $z = 0.373$) when testing the difference in proportions.

Continuing with our analysis of player A’s choices, **Hypothesis 2** predicts that player A will be more conservative when receiving support from AI. Again, we pool treatments based

⁹ $X^{\text{No AI}} = \{\text{Baseline, Only A}\}$ and $X^{\text{AI}} = \{\text{Only B, Both}\}$.

on access to AI and find that player A chooses ‘In’ 40% of the time with and without AI support. Thus, we fail to reject the null (effect size = 0%).

When neither player has AI, player B chooses ‘Roll’ 70% of the time. This result remains mostly unchanged in treatments where player B receives AI assistance. Sessions where only player A receives AI assistance stand out with a roll rate of 50% compared to 68.9% across the other three treatments (effect size = 18.9%, $p = 0.062$; $z = 1.869$). With access to AI being public information within each group, this suggests the knowledge that only player A will receive guidance from an AI assistant may influence player B’s decision. Recall that our sample sizes would reliably detect differences only of roughly 25 percentage points or larger. This helps explain why the evidence is suggestive but not decisive: effects in the moderate range are plausible given the current precision.

We refer to the strategy profile (‘In’, ‘Roll’) as the *cooperative outcome* as it constitutes the greatest expected collective payoff for each pair. In the benchmark treatment, we observe the cooperative outcome in 20% of pairs. In sessions where only player A receives AI assistance, the proportion of cooperative outcomes decreases to 16.7%, indicating minimal impact. Conversely, there is modest evidence that player B’s access to AI improves cooperation. In treatments where player B has the option to use AI – Only B and Both – we observe the cooperative outcome in 33.3% and 30% of pairs, respectively. Pooling observations by player B’s access to AI, we find a 13.34 percentage point increase in cooperative outcomes when player B receives AI assistance (effect size = 13.34%, $p = 0.093$; $z = 1.687$). This indicates that while large effects are ruled out, moderate effects cannot be excluded. Taken together, the disconnect between individual choices and pair-wise choices across treatments indicates AI assistance may not significantly impact individual decisions; rather, it may help to coordinate cooperative trustors with cooperative trustees. This provides directional support for **Hypothesis 5**.

We now turn to the impact of AI assistance on beliefs. First-order beliefs (τ_A) represent player A’s confidence that player B will choose ‘Roll’. Second-order beliefs (τ_{BA}) reflect player B’s perception of player A’s beliefs. To provide a more intuitive interpretation of results where higher values correspond with increased trust, we map qualitative responses from the

Certainly Choose Don't Roll	→	0
Probably Choose Don't Roll	→	0.25
Unsure	→	0.5
Probably Choose Roll	→	0.75
Certainly Choose Roll	→	1

Table 1: Numerical values assigned to elicited beliefs.

post-experiment survey to numerical values according to Table 1.

Consistent with existing literature, our findings confirm that beliefs and behavior are closely interconnected. Specifically, we observe that A is more inclined to choose ‘In’ when they are confident B will select Roll. As selecting ‘In’ report an average τ_A of 0.74, translating to a belief that B will “probably choose Roll.” On the other hand, those choosing ‘Out’ exhibit reduced trust in B, with a lower average of 0.44. Moreover, Bs who decide to ‘Roll’ have an average second-order belief of 0.73, while those who choose ‘Don’t Roll’ show a significantly lower average τ_{BA} at 0.37.

Figure 4 presents average first- and second-order beliefs across the four treatments. In the benchmark treatment, trustors display an average τ_A of 0.52, indicating a neutral level of trust. The inclusion of AI assistance for B increases this average to 0.61. However, this pattern does not persist across the Only A and Both treatments. When pooling observations by B’s access to AI, we find an average τ_A of 0.529 in treatments without access, only slightly less¹⁰ than when B is assisted by AI. In line with the patterns observed in choices, there is minimal evidence that AI significantly influences A’s beliefs.

Second-order beliefs appear to be equally inconsistent across treatments, except in cases where only A receives AI assistance. In such cases, B’s average τ_{BA} is 0.475, markedly lower than the 0.629 observed across other treatments (effect size = 0.154, $p = 0.031$; $t = 2.222$). This divergence might be attributed to the public information that only A can access AI, leading B to form a more pessimistic perception of A’s beliefs. It is also consistent with the low Roll rate in the only A treatment compared with the others.

¹⁰A difference of 0.07.

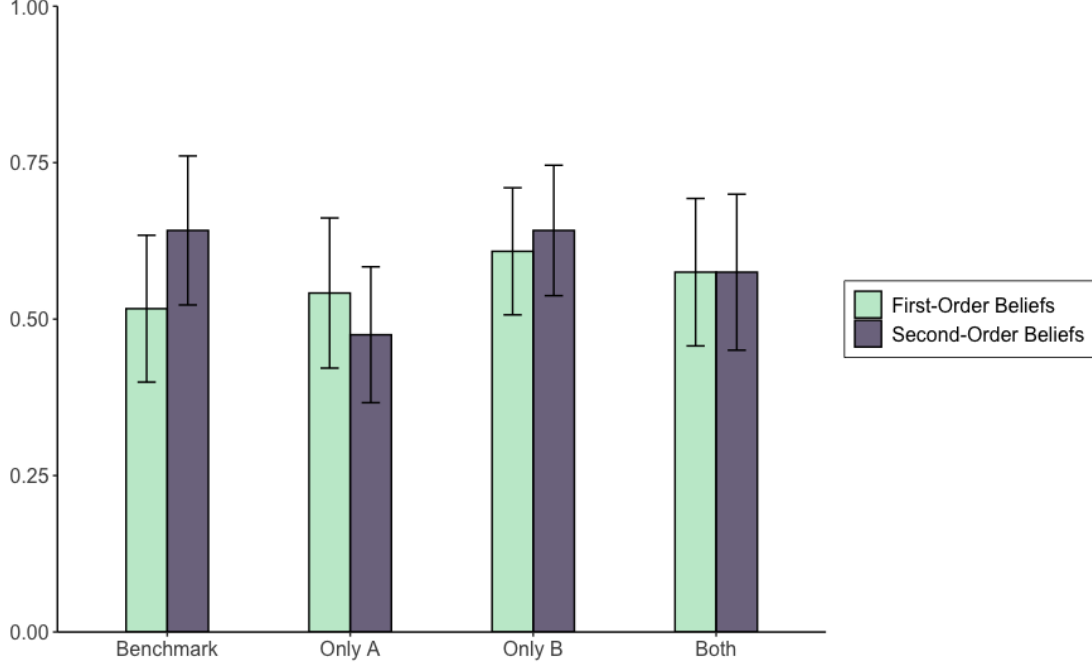


Figure 4: Average first-order and second-order beliefs across the four treatments. Confidence bands are calculated at the 95-percent level.

3.2 Message Classification

3.2.1 Player B

We classify messages sent by player B according to the *type* of message sent and the *method* by which it was sent. The type of message that player B sends takes one of the following values: ‘promise’, ‘asking’, ‘empty’, ‘skip’, ‘fairness’, ‘anti-promise’. Meanwhile, we classify the *method* into one of ‘Own’, ‘AI’, ‘Mixed’, ‘Skip’. Each of these classifications is briefly discussed below, with further details in the appendix.

Message Type

Message type primarily concerns the *content* of the message. We classify player B’s messages using a set of theory-informed categories motivated by the role of pre-play communication in trust games. Prior work emphasizes that statements about one’s own intended action (especially an explicit intention to reciprocate) are the most direct form of non-binding commitment and are therefore central for belief formation and trust decisions. Guided by this idea, we pre-specify a coding scheme that separates messages in which player B states

an intention about their own future move from messages that only recommend what player A should do or appeal to an outcome norm.

We dissect the message sent by player B into what player B indicates about *player A's* potential move and what player B expresses about their *own* intended move. For each of these components, we assign a tertiary label. For the first component – the piece of the message involving player A's move – we assign a label from $\emptyset/In/Out$, with $\emptyset$ representing no definitive information conveyed. Similarly, for the second component – on player B's own move – we assign a label from $\emptyset/Roll/Don't\ Roll$. This categorizes any message sent by either player B or the AI into one of 9 pairs. This codification allows us to assign labels to the messages according to Table 2. Note that this assignment is for explicit messages; when player B opts not to send a message, they are assigned the message type “skip”. Further details of message-code assignment are left to the appendix.¹¹

Msg Vec	Label
$(In, Roll)$	<i>Promise</i>
$(\emptyset, Roll)$	
$(Out, Roll)^*$	
$(In, \emptyset)$	<i>Asking</i>
$(\emptyset, \emptyset)$	<i>Empty</i>
$(Out, \emptyset)$	<i>Fairness</i>
$(In, Don't\ Roll)$	<i>Anti-promise</i>
$(\emptyset, Don't\ Roll)$	
$(Out, Don't\ Roll)$	

Table 2: Each message sent by player B (or recommended by AI) is encoded as vector which captures what player B intends to do and what they propose player A should do. Note that (Out, Roll) does not constitute a cooperative outcome unlike the other Promises. We nonetheless include it in Promise since it demonstrates Player B's intention to play Roll. We do not observe any encoded message of (Out, Roll).

We are chiefly interested in how player B expressing an intention to play ‘Roll’ affects choices, outcomes, and beliefs. We therefore code a “promise” as any explicit statement of intent by player B to play ‘Roll’. Symmetrically, if player B explicitly states an intention to play

¹¹Additionally, a visualization of the transformation from player B's initial message through the AI to their final message can be found in Appendix Figure A.1, with additional examples following in panels (a)-(e) of Figure A.2.

‘Don’t Roll’, we code it as an “anti-promise.” This category matters because it directly signals low trustworthiness. If player B does not state their own intended move but asks player A to play ‘In’, we code the message as “asking.” If player B instead recommends that player A play ‘Out’, which guarantees an equal payoff for both players, we code the message as “fairness.” Unlike promises and anti-promises, “asking” and “fairness” do not state B’s intended move. Because A’s choice ultimately depends on her belief about B’s likelihood of ‘Roll’, these messages are typically less informative and may be less effective than intention statements. Even so, these messages may still coordinate behavior to some extent by shaping player A’s expectations about what player B is trying to achieve (e.g., encouraging ‘In’ or proposing the safe equal-outcome option). Finally, if no clear intention or recommendation is communicated, we assign the label “empty.”¹²

A breakdown of message types for non-skipped messages across all treatments is provided in Figure 5. The largest share (41%) of sent messages are Promises, with 90% of sent messages being comprised of Promises, Asking, and Empty messages.

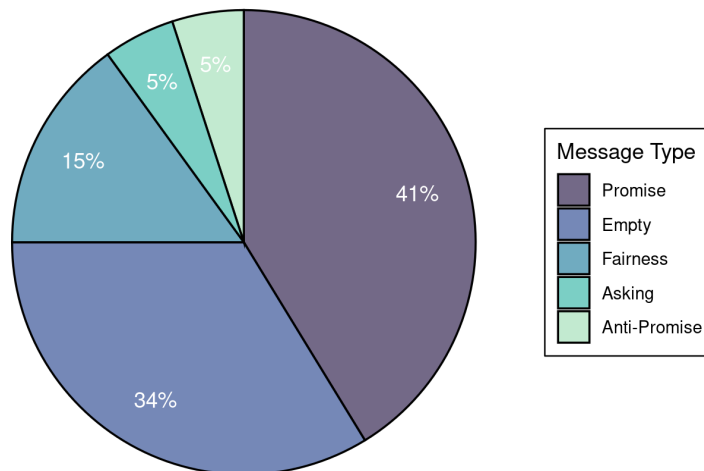


Figure 5: Types of messages sent across all treatments, omitting skipped messages. Note that $\approx 1/3$ of B’s in the sample opted to skip sending a message.

¹²Note that no truly “empty” messages are sent: all messages sent contain *some* content. Therefore, it may be helpful to think of the “Empty” label as “Junk” based on the complement set of messages already described.

Figure 6 illustrates the distribution of player choices based on player B’s message type. Messages that assure player B will choose ‘Roll’ or ask player A to choose ‘In’ result in the highest frequency of A choosing ‘In’. We expect messages classified as ‘empty’ to elicit similar responses from A as cases where B opts out of sending any message, as both situations lack any substantive signal of B’s intentions or trustworthiness. However, our data show trustors choose ‘In’ more frequently when receiving an empty message (37.03%) than no message (25%). Several of the empty messages are disconnected from the experiment itself but contain relatively positive language.¹³ It may be the case that sending a positive message can help to establish trust even if the message is irrelevant to the game.

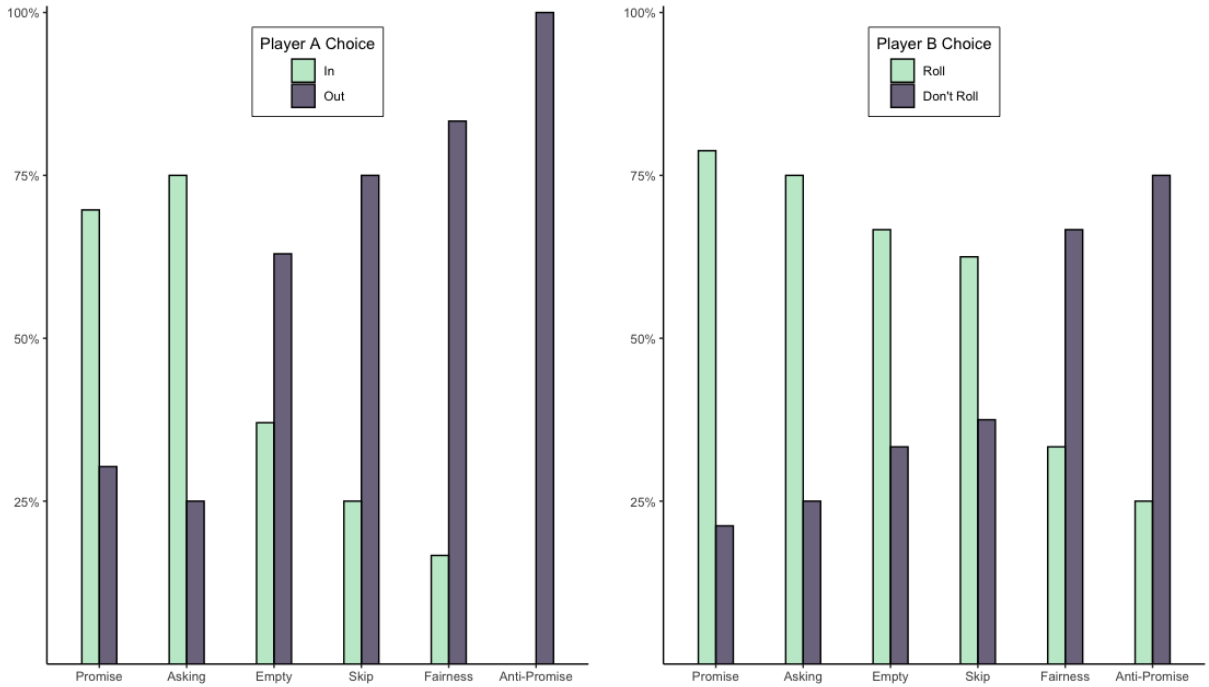


Figure 6: Distribution of player choices by message classification.

Messages categorized as ‘fairness’ and ‘anti-promises’ share similar purposes but differ in the signals they convey. Fair messages explicitly encourage A to choose ‘Out’ by presenting it as the safest option, while anti-promises discourage choosing ‘In’ by disclosing B’s intent to choose ‘Don’t Roll’. As anticipated, both types of messages elicit the lowest in-rates across all pairs. Notably, no trustors chose ‘In’ after receiving an anti-promise, underscoring the

¹³For example, one message contained “Hey, I hope we have a good game (:”

strong deterrent effect of such messages.

Hypothesis 3 postulates that Player B will send more promises when they have access to an AI assistant. Our data show a 75% increase in the proportion of messages containing promises when B gains access to AI (effect size = 75%, $p = 0.067$; $z = 1.84$). Expanding the notion of a promise to include ‘asking’ increases statistical significance to the 97-percent level.¹⁴ Specifically, our 95-percent confidence interval indicates the presence of AI for Player B yields a 1.83 to 34.8 percentage point increase in messages categorized as ‘promise’ or ‘asking’. Figure 7 displays these findings visually.

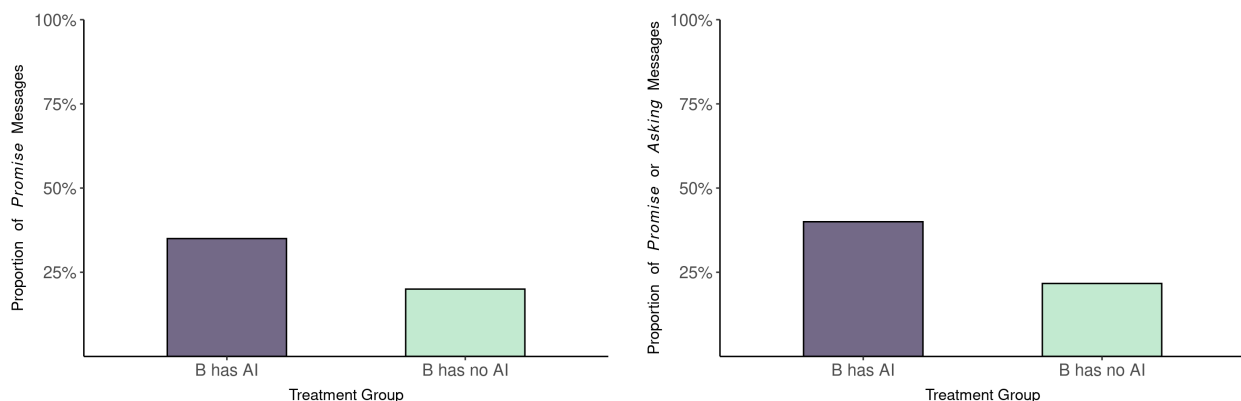


Figure 7: Left panel: proportion of messages sent by player B which are promises. Right panel: proportion of messages sent by player B which are promises or asking.

Message Method

As opposed to message type, which concerns the content of the message, message method concerns the *authorship* of the message. Since player B may edit the message suggested by the AI before sending it to player A, we aim to discern whether the content of the message was primarily dictated by AI, by the agent, or by a reasonable mix of the two. We use a normalized version of the Levenshtein edit distance (Levenshtein et al., 1966) to determine the pairwise relative distances between player B’s first message, the AI’s suggested message, and the actual sent messages¹⁵. When plotting the normed Levenshtein distance between

¹⁴We consider ‘asking’ alongside ‘promise’ as to include any message which directly suggests that player A play ‘In’, opening up the possibility for a cooperative outcome.

¹⁵The Levenshtein distance is a metric which reports the total number of single-character edits needed to transform one string into another. In particular, the distance measures the number of insertions, deletions, and substitutions required to transform one of its inputs into the other. We implement a normed version of this metric, which scales the traditional Levenshtein distance between two strings by the length of the larger

the the first and sent messages against the AI-suggested and sent messages (Figure A.3 in the appendix), a clear grouping structure can be seen. To verify this grouping structure, we implement a k -means cluster classification with $k = 3$ means¹⁶ to produce the labeling assignment. Each member of the research team independently inspected each message to ensure accurate labels. The upshot is that sent messages which are labelled ‘Own’ have near-identical similarity to the first messages player B sent compared to the AI’s suggested message; sent messages labelled ‘AI’ have near-identical similarity to the AI’s suggested message compared to the first message player B sent, and ‘Mixed’ messages player B sent are those which bare a fair similarity to both the first and the AI-suggested message. Further details can be found in the appendix.

Figure 8 shows the breakdown of how B players sent their messages, provided that they sent one at all. The majority of sent messages are primarily their own compositions, with near equal shares of messages being crafted entirely by the AI or a mix of AI and player B. It should be noted that both Figure 8 and Table 3 are restricted to treatments when player B has an AI assistant.

Further examination of the effect of AI’s usefulness in message composition reveals that players who spend time working with, as opposed to relying upon, the AI assistant received the best results as measured by convincing player A to play ‘In’. Figure 9 shows the interaction between time spent creating their message and the method used to create their message. When players spent additional time crafting their message (a proxy for effort) they are more successful in soliciting ‘In’ from player A. This effect was magnified when the player incorporated the feedback from the AI assistant into their message thereby creating a ‘mixed’ message. However, the same positive effect did not apply to those who relied entirely upon the AI to create the message they sent to player A, if anything the performance of the message decreased over time spent on the AI message.

In contrast, Player A was less likely to comply with Player B’s request the longer they waited

string. This transforms the metric into a measure of similarity between the two strings lying between 0 and 1, as the maximum length of the two input strings is exactly the maximum number of single-character edits needed to transform one string into the other.

¹⁶See Appendix A.2.2 for details.

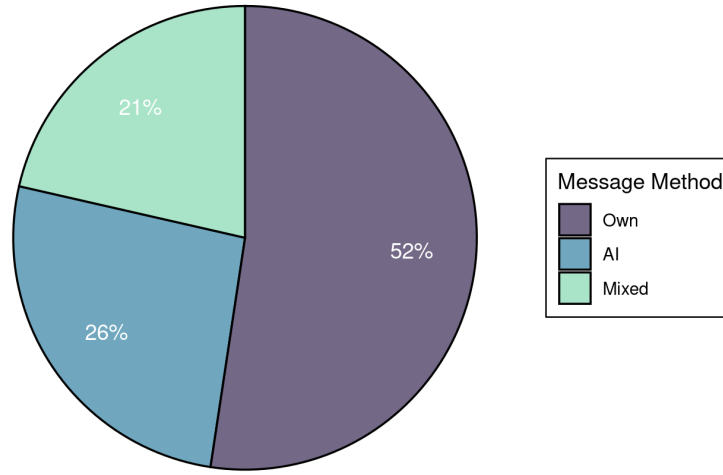


Figure 8: Over half of player B's who sent a message did so using mostly their own authorship. Details of how these methods were imputed are included in the appendix.

for the message, as shown in Figure 10. As the wait time increased, Player A was more inclined to choose 'In' when Player B asked them to play 'Out,' and more likely to choose 'Out' when Player B requested 'In.'

Figure 11 shows the distribution of player choices by method across the four treatment groups. Without AI assistance, Player B can either opt not to send a message or compose their own. Consistent with earlier results, player A selects 'In' at a relatively low rate of 25% in the absence of any message. When the treatments allow Player B to send messages that are either partially or completely generated by AI ('Only B' and 'Both'), these AI-assisted messages result in unexpectedly lower 'In' rates: 44.4% for 'mixed' messages and 36.4% for fully AI-authored messages, in contrast to 54.5% for original messages. These findings indicate that the authorship of the message may not influence Player A's decisions as much as the actual content of the message.

Combining message type with message method, Table 3 displays raw counts of the types of messages player B sent and how they sent them.

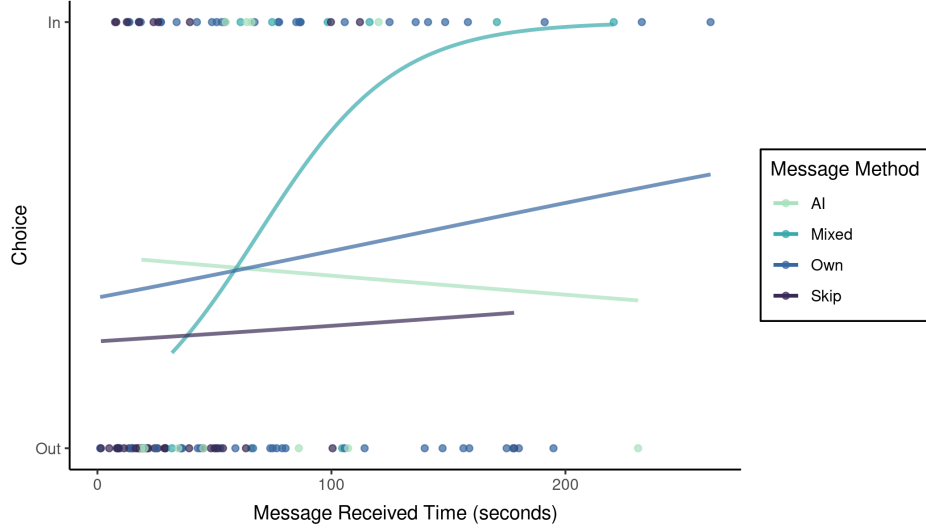


Figure 9: Comparison of Player A’s choice versus the time in seconds they waited for Player B’s message, categorized by Player B’s message delivery method. Without an interaction term, time spent waiting significantly increased the probability ($p = 0.03$) that players would choose ‘In’.

Message Type	Own	Mixed	AI	Total
Promise	9	6	6	21
Asking	0	1	2	3
Empty	9	1	1	11
Fairness	4	1	0	5
Anti-Promise	0	0	2	2
Total	22	9	11	42

Table 3: Summary of the number of non-skipped messages sent by player B according to their type (row) and method (columns). Treatment is restricted to cases when player B has an AI.

To test **Hypothesis 4**, we group messages containing a promise to choose ‘Roll’ according to whether they are authored by Player B or suggested by AI. Player B fulfills their own promises 85.7% of the time, compared to 40% for AI-suggested promises. While this difference is not statistically significant (effect size = 45.7%, $p = 0.137$; $t = 1.80$) and the 95% confidence interval is wide (0.3% – 91.1%) due to small sample sizes, the magnitude of the observed effect suggests that AI-suggested promises may lower the perceived cost of breaking a promise. These findings are therefore best interpreted as exploratory, providing preliminary evidence of a behavioral pattern that warrants further investigation.

There is a negative relationship between the time it took Player B to send their message and choosing which action to play and their choice between ‘Roll’ and ‘Don’t Roll’, as shown in

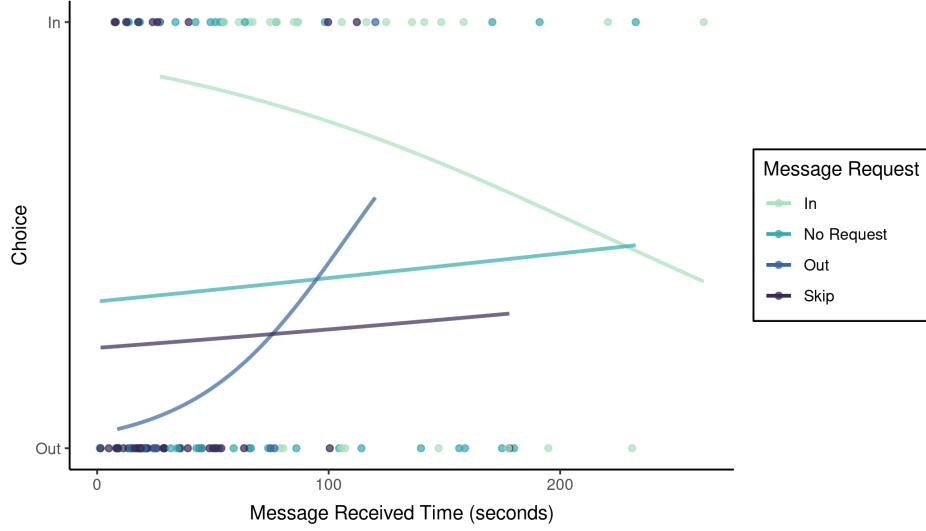


Figure 10: Comparison of Player A’s choice versus the time in seconds they waited for Player B’s message, categorized by the choice Player B requested Player A makes.

Figure 12. The longer it took for Player B to make their decision after sending a promising message that they would play ‘Roll’, the less likely they were to follow through with their promise.

3.2.2 Player A

Message classification for player A’s AI is naturally less intensive:¹⁷ when player A has an AI, we classify the interpreted message on behalf of the AI¹⁸ as either “no clear suggestion”, “strongly advises playing ‘In’”, “weakly advises playing ‘In’” and “primarily advises playing ‘Out’.”¹⁹ In the appendix, we collapse the strong/weak ‘advise-In’ into a single label for a symmetric assignment. Furthermore, explicit examples of player B’s sent message and the corresponding interpretation from player A’s AI can be found in panels (a)-(e) of Figure A.6 in the appendix.

Figure 13 shows the suggestions made by player A’s AI compared to the choice which player A ultimately made in the game. Proportionally, it seems that player A closely follows the advice

¹⁷Indeed, much of the time, A’s AI addresses player B’s message, reviews the potential outcomes of the game, and advises player A to play according to their own risk preferences.

¹⁸This was done by each member of the research team independently by hand for all messages, and these independent labels were compared and contrasted until unanimity was reached for each message label.

¹⁹Our data on A’s AI messages is absent from the notion of a ‘strong’ v. ‘weak’ suggestion of ‘Out’.

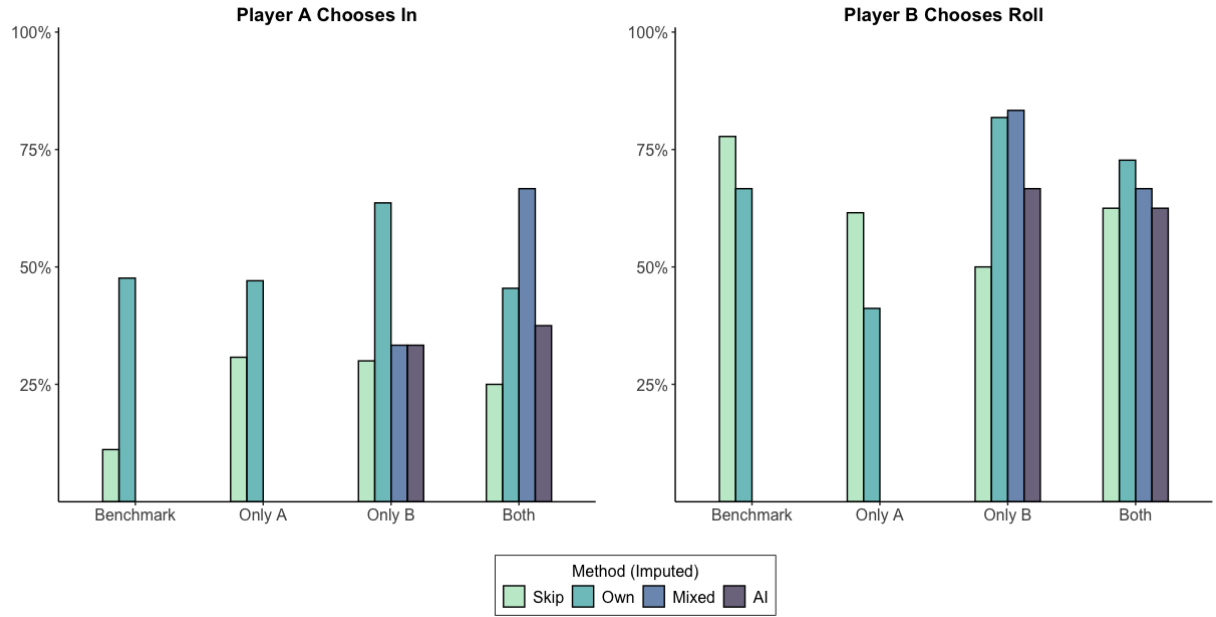


Figure 11: Distribution of player choices across treatments by message authorship.

of their AI assistant when the assistant suggests ‘Out’. On the other hand, this suggestion is the least frequent of the three in the sample, with only 18% of AI suggesting that player A chooses out.

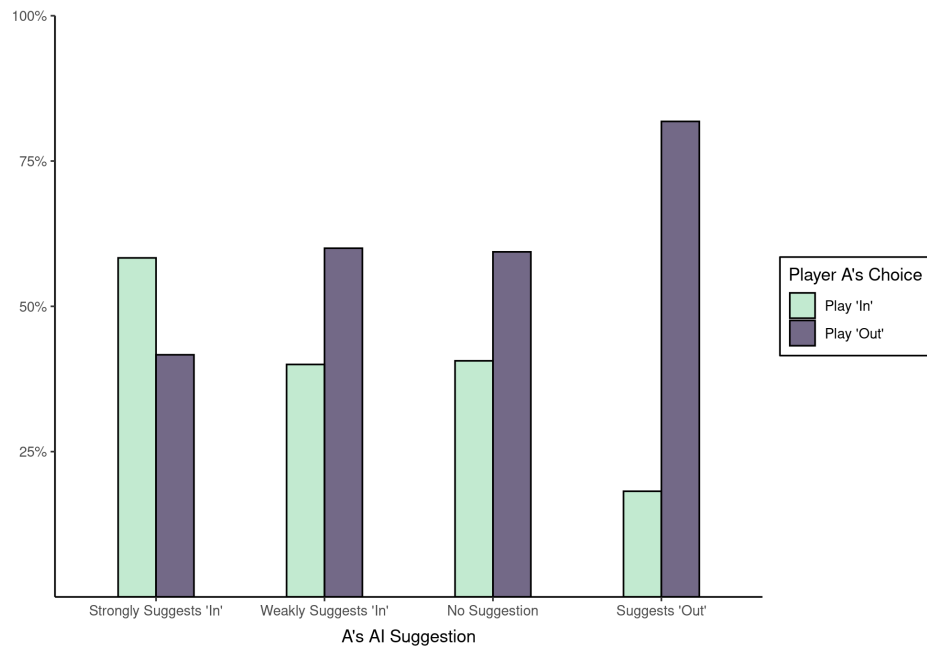


Figure 13: Player A's choice and the associated interpretation by player A's AI.

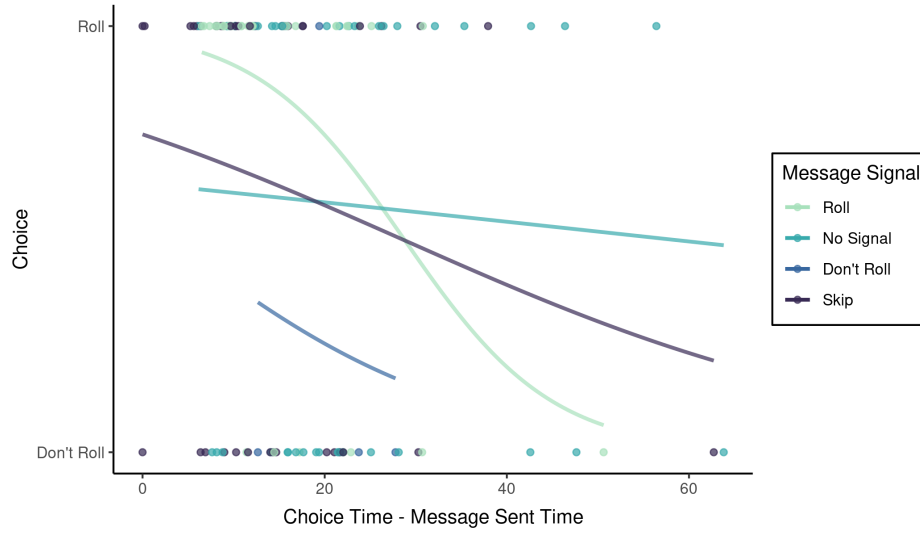


Figure 12: Comparison of Player B's choice versus the time they took to make their choice after sending their message, categorized by Player B's signal of what choice they would make.

Why do we see such a small proportion of AI suggesting 'Out'? Figure 14 displays the suggestions made by player A's AI, this time alongside the type of message that player B sent. Recall that an assignment of 'fairness' indicates that player B made no mention of their own move, rather they simply request that player A choose 'Out'. When player B sends a message of this type, player A's AI is almost guaranteed to advise similarly.

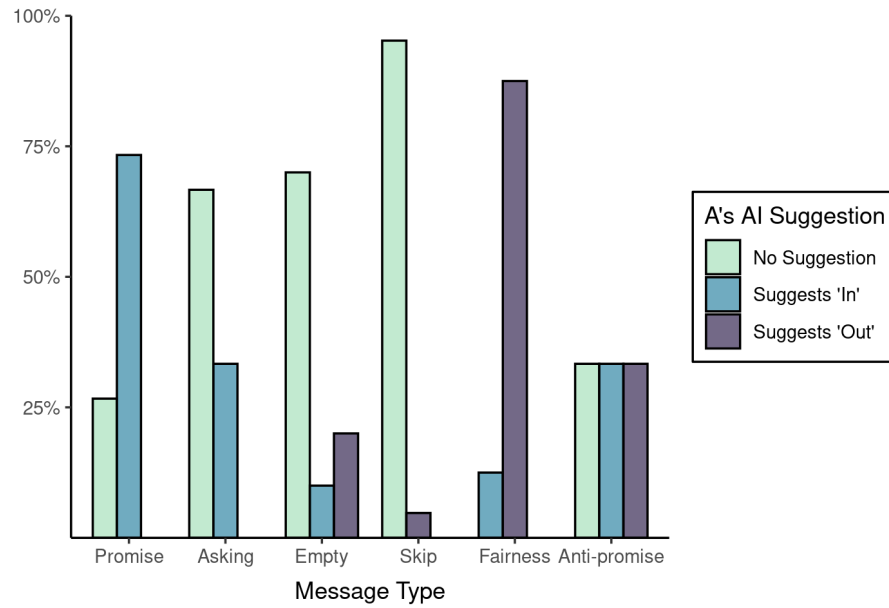


Figure 14: Player B's message type and the associated interpretation by player A's AI.

3.3 Risk

We elicit subjects’ risk preferences via an unincentivized five-question Holt-Laury quiz (hereby *HL*). This quiz is an industry standard for evaluating an individual’s degree of risk aversion. The HL survey presents five rows of binary choices over lotteries. In our HL implementation (shown in A.5.3), the payoffs stay fixed while the probabilities associated with earning these payoffs vary across rows. The lottery on the LHS is generally more balanced than the lottery on the RHS, e.g. \$120/\$80 on the LHS versus \$190/\$5 on the RHS. As the subjects work down the survey, the RHS becomes progressively safer, while the LHS becomes relatively riskier.

A rational agent with consistent risk preferences should not switch back to the LHS lottery after selecting the RHS lottery. Agents who do not perform such a switch back are said to satisfy the *single-switching* property. Among $n = 120$ subjects under the player A role, 104 of them satisfied single-switching.²⁰

We label the first row in which a player switches from the LHS to the RHS as their “switch” point, or *SwP*. An SwP of 1 means the player exclusively chooses the riskier lottery (which we do not observe in our study), while an SwP of 6 denotes a player who exclusively chooses the safe option (i.e. never switches). To reduce confusion, we map the 5 observed switch points to a discrete risk aversion rank, or *RAR*. In this classification, a rank of 1 (SwP= 2) denotes the least risk-averse (most risk-loving) subjects, while a rank of 5 (SwP= 6) denotes the most risk-averse subjects.

Among those player A subjects with rational risk preferences,²¹ the distribution of RAR is shown in Table 4, along with the average first-order (τ_A) belief of such players and the proportion of them who chose “In”. RAR values go from least to most risk-averse. Similar tables, organized by choice and trust level, can be found in Appendix A.6.1.

We use logistic regression to analyze how the role of trust compares to that of risk in

²⁰Although we elicit subject-level risk aversion via a HL survey regardless of role, Player B does not face significant risk when making their choice, since they receive the same payout regardless of chance.

²¹Note that we are only able to assign a rank to player A’s risk preferences if they satisfy the single-switching property.

	RAR	Total (n)	Prop. ‘In’	Average Belief (τ_A)
<i>Least risk-averse</i>	1	1	1.000	0.750
	2	13	0.385	0.750
	3	43	0.372	0.541
	4	24	0.375	0.427
<i>Most risk-averse</i>	5	23	0.435	0.598

Table 4: Distribution of player A subjects’ risk aversion level, assuming they satisfy the single-switching property on a Holt-Laury quiz. We do not observe subjects who always choose the riskier lottery, so we rank the 5 remaining switch points corresponding to risk aversion level. $\tau_A = 1$ indicates player A certainly trusted that player B played ‘Roll’, while $\tau_A = 0$ indicates that player A certainly thinks player B chose ‘Don’t Roll’.

influencing player A’s behavior. Specifically, we estimate²²

$$\text{logit}[P(\text{In} = 1)] \sim \beta_0 + \beta_1 R + \beta_2 B + \beta_3 (R \times B),$$

where R and B denote the sample-relevant, demeaned²³, RAR and τ_A variables, respectively. Thus, coefficient estimates can be interpreted with respect to average RAR and τ_A levels. We cluster standard errors at the session level. Regression estimates are shown in Table 5.

Coefficient	β Est.	SE	t-stat	p-value
Intercept	-0.522	0.206	-2.536	0.029 *
RAR	-0.411	0.984	-0.418	0.685
τ_A	3.872	0.801	4.833	0.001 ***
$\text{RAR} \times \tau_A$	8.347	4.343	1.922	0.083 .

Table 5: SEs clustered at session-level. Estimates are demeaned: the intercept is interpreted to be for those with average risk-aversion and trust levels across those player A’s with rational risk preferences.

The intercept captures the baseline log-odds of choosing ‘In’ for a subject with average risk preferences and average beliefs about player B’s move. Such a player²⁴ selects ‘In’ about 37% of the time.

We find that beliefs are highly statistically significant ($p = 0.001$), demonstrating that player A’s trust in player B plays a non-zero role in predicting player A’s behavior. Assuming player

²²Recall that the “logit” is equal to the log-odds of a successful event. In general, this amounts to a regression of the form $p(x) = (1 + \exp(-(\beta_0 + \beta_1 x)))^{-1}$, or, so that the left-hand side is linear, $\ln \frac{p(x)}{1-p(x)} = \beta_0 + \beta_1 x$.

²³That is, the variable minus it’s own sample mean when restricting the sample to player A’s with rational risk preferences.

²⁴That is, a mean-representative player A with a RAR of 3.529 and τ_A of 0.555 for our restricted sample.

A has average risk preferences over our sample, increasing from the bottom quartile of trust in player B to the top quartile of trust in player B is associated with a 40.4% increase in the likelihood of player A selecting ‘In’. On the other hand, we fail to reject the null hypothesis that risk preferences play an insignificant role in determining whether player A chooses ‘In’ or ‘Out’, by themselves.

The positive interaction between beliefs and risk indicates that beliefs translate more strongly into trusting behavior for more risk-averse subjects. In other words, the impact of beliefs on trusting behavior increases with risk aversion. To see this, we will consider those with relatively extreme risk preferences and beliefs, as defined by having a RAR or τ_A in the bottom or top quartile of the sample. We label those in the bottom quartile as “low” (L) and those in the top quartile as “high” (H), and we consider the differenced effects derived from the estimated logit model. Table 6 reports the change in the expected probability²⁵ of choosing ‘In’ associated with a rise in beliefs from the bottom quartile to the top quartile of the sample, evaluated at different levels of risk aversion.

Risk	Est. $\Delta P(\text{‘In’})$ low $\tau_A \rightarrow$ high τ_A	SE
L	0.306	0.087 ***
H	0.479	0.072 ***

Table 6: Effect of moving from relatively low trust in player B to relatively high trust in player B on the predicted probability of selecting ‘In’, for relatively low and high levels of risk-aversion. Here, ‘relatively’ means in the bottom (low) or top (high) quartile of the sample. Predicted probabilities are computed using the estimated logit model. Effects are therefore in percentage points for percentages in $[0, 1]$. SEs are based on CR2 cluster-robust standard errors at the session level.

Increasing beliefs from low to high increases the probability of choosing ‘In’ by 30.6pp for the most risk-tolerant subjects, and by 47.9pp for the most risk-averse subjects. Thus, beliefs play a larger role for the more risk-averse subjects in our study.

Figures 15 and 16 visually summarize the results from this logit estimation. Figure 15 highlights beliefs impacting choices more dramatically for more risk-averse subjects. In Figure 16, we see that changing beliefs impacts the predicted probability of choosing ‘In’ for fixed levels of risk far more than changing risk does for fixed levels of belief. Furthermore,

²⁵Exact probabilities for four different combinations of risk and trust levels can be found in Table A.5 of the appendix.

relatively risk-averse subjects are impacted by changes in trust more dramatically than their risk-tolerant counterparts.

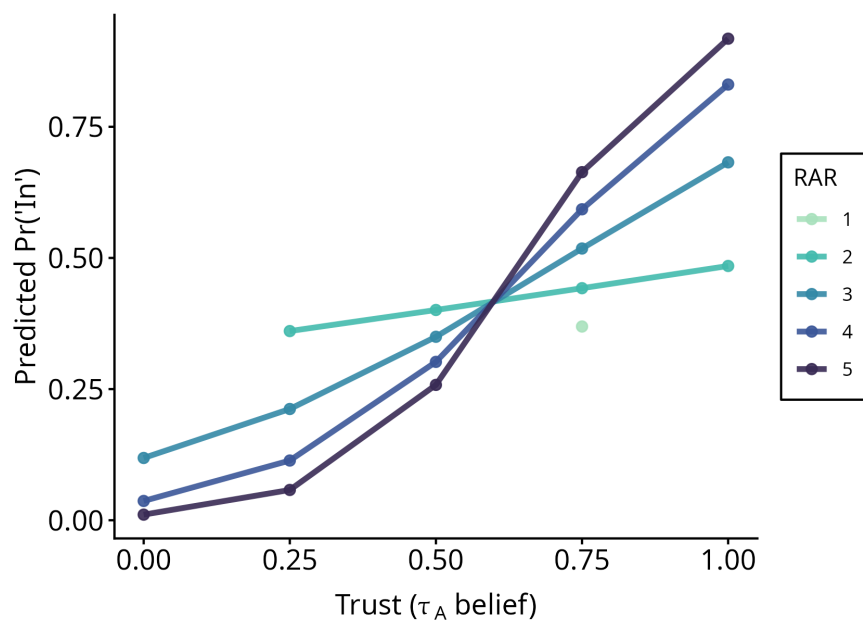


Figure 15: How trust impacts the logit-predicted probability of player A selecting 'In', for different levels of risk aversion.

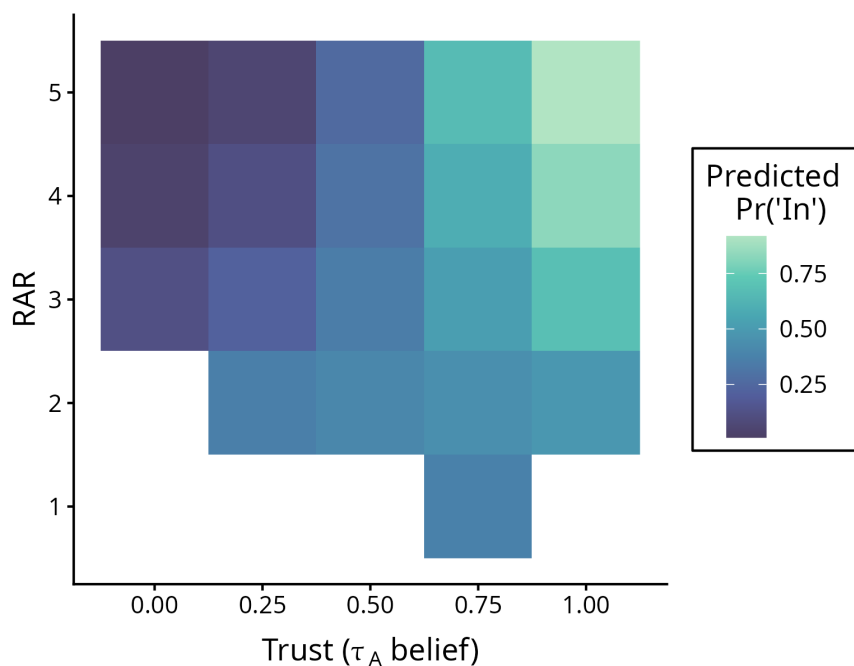


Figure 16: The interaction of risk and trust on the logit-predicted probability of player A selecting 'In'. Higher values on the vertical axis indicate more risk-averse subjects, while rightward values on the horizontal axis indicate player As who are more trusting that player B chose 'Roll'.

We conclude that player A’s beliefs impacted their behavior far more than their risk attitude. This, in essence, is the *trust* involved in the game: player A’s beliefs, as influenced by player B’s message, drive player A to make their investment over their own risk priors.

4 Discussion

4.1 Summary of Hypotheses and Results

Table 7 summarizes the main hypotheses, the outcome measures, and the results from the statistical tests reported in the Results section.

Hyp.	Prediction (plain language)	Comparison	Result
H1	If B has AI, A is more willing to choose <i>In</i> .	Proportion of A choosing <i>In</i> ; pooled comparison $X^{AI} = \{\text{Only B, Both}\}$ vs. $X^{NoAI} = \{\text{Baseline, Only A}\}$.	41.67% vs. 38.33%, $p = 0.709$, $z = 0.373$ (not significant).
H2	If A has AI, A becomes more conservative (less <i>In</i>).	Proportion of A choosing <i>In</i> ; pooled comparison by A’s AI access.	40% vs. 40% (no change).
H3	If B has AI, B is more likely to make a promise (e.g., “I will Roll”).	Proportion of B messages that contain a promise (and in an expanded definition, “promise or asking”); comparison by B’s AI access.	Promise messages increase by about 75% when B has AI, $p = 0.066$, $z = 1.840$ (suggestive).
H4	If the promise is suggested by AI, B is less likely to keep it.	Promise follow-through (kept vs. broken); compare AI-suggested promises vs. otherwise.	Follow-through lower when AI suggested (40% vs. 85.7%), $p = 0.137$ (inconclusive).
H5	AI increases cooperative outcomes by improving coordination.	Cooperative outcome (‘In’, ‘Roll’); pooled comparison by B’s AI access.	Increase of 13.34 pp when B has AI, $p = 0.092$, $z = 1.687$ (suggestive).

Table 7: Hypotheses and Results

4.2 Contributions and Implications

The main contribution of our findings is to show that AI can affect trust outcomes at the pair level, through a suggestive increase in coordination between trustors and trustees, rather than by simply changing one side’s behavior in isolation. Across treatments, we do not find strong evidence that AI access by itself substantially shifts the trustor’s propensity to choose ‘In’. Instead, the key pattern is at the pair level: when the trustee has access to AI, the probability of reaching the cooperative outcome (‘In’, ‘Roll’) rises, though the effect falls short of statistical significance..

This finding extends the pre-play communication literature by examining what happens when AI assists the trustee in composing the pre-play message. [Charness and Dufwenberg \(2006\)](#) find that communication increases cooperation and that this effect is driven by messages categorized as promises. In our experiment, we find preliminary evidence that trustee-side AI assistance increases both the frequency of promise messages and the frequency of the cooperative outcome. At the same time, we do not find clear evidence that trustee-side AI assistance increases Player A’s belief that Player B will choose ‘Roll’, Player A’s likelihood of choosing ‘In’, or Player B’s likelihood of choosing ‘Roll’. Our results therefore suggest that AI assistance may affect promise-making and pair-level coordination without significantly changing average trust and trustworthiness.

This distinction is highly practical: in many real-world strategic interactions such as borrower–lender relationships, contracting, and partnership formation, trust ultimately hinges on whether expectations line up with subsequent actions. This is consistent with organizational and IS conceptions of trust as reliance under uncertainty ([Mayer et al., 1995](#), [McKnight et al., 2002](#)). AI may therefore generate welfare gains by improving pair-level coordination even when it does not significantly change average beliefs or individual choices.

A direct implication of the results presented in this study is that AI may be most useful in settings where cooperative failures commonly come from misaligned expectations rather than from fixed preferences for opportunism. In our design, AI is associated with a non-conclusive increase in the frequency of explicit commitments in pre-play communication

(promise-making), which may help trustors and trustees form more similar expectations about what will happen next. When expectations are better aligned, the interaction is less likely to end in the “near-miss” outcomes that reduce welfare: a trustor holding back even though the trustee would have followed through, or a trustor trusting when the trustee does not reciprocate. Because we do not observe whether AI improves message clarity or changes Player A’s interpretation of Player B’s message, clearer communication and interpretation remain possible explanations for this pair-level pattern rather than established mechanisms.

At the same time, our promise-follow-through results highlight a potential concern: if AI makes it easier to produce commitment language, then the gap between promising and delivering may widen in some contexts. While our evidence on this point is not statistically decisive in the current sample, the directional difference in promise fulfillment suggests that the origin of a promise may matter for whether it is subsequently honored. This possibility is consistent with promise-keeping accounts, under which a self-authored promise may create a stronger personal obligation to follow through (Vanberg, 2008). Because the difference is statistically inconclusive and we have no measure for this sense of obligation, we treat this explanation as tentative.

Our research also adds to the discussion on algorithm aversion and trust in AI (Glikson and Woolley, 2020). The low number of subjects directly adopting the AI-suggested messages in our experiment is consistent with evidence that people may resist algorithmic advice (Dietvorst et al., 2015). However, our experimental design does not allow us to determine whether the low adoption of AI-recommended messages is due to aversion to AI or greater confidence in one’s own judgment.²⁶ As Logg et al. (2019) point out, individuals tend to be more receptive to algorithmic advice in areas where there is a clear and measurable external standard of accuracy. In contrast, the quality of suggestions for interpersonal communication is less easily quantified, which may lead participants to rely more heavily on their own judgment. Accordingly, the low adoption rate should not be interpreted as direct evidence of algorithm aversion.

²⁶Our post-experiment survey elicited overall trust in AI from the subjects. However, we did not find evidence that subjects who chose not to adopt AI-recommended messages were more averse to AI than other subjects.

Relative to studies in which AI provides information, strategic advice, or deception-detection support (Harris et al., 2023, Bai et al., 2023, Serra-Garcia and Gneezy, 2023), our results extend the literature by examining AI’s role in producing and interpreting a voluntary message between two humans before a trust decision. Methodologically, the experiment also shows how a classic trust-game paradigm can separate trustee-side AI assistance in message production from trustor-side AI assistance in message interpretation.

4.3 Limitations and Future Research Directions

Our findings should be interpreted with several limitations in mind.

First, AI access is *common knowledge* in our design: participants know whether their counterpart is assisted by AI. As a result, our treatment effects reflect the impact of AI in an environment with disclosure, where AI involvement is salient to both sides. This is a limitation because any difference we observe can combine (i) changes in the content generated with AI assistance and (ii) changes in how participants respond when they know AI is involved, including possible initial skepticism toward AI-assisted communication. Our design therefore cannot cleanly separate content effects from disclosure/salience effects.

Second, some of the main treatment differences are directionally consistent with our hypotheses but lack traditional statistical strength. This is consistent with limited power for certain comparisons (especially those that condition on relatively infrequent events, such as AI-suggested promises and subsequent promise keeping). The experiment includes 30 pairs per treatment, and the minimum detectable effect for our primary two-group comparisons is approximately 0.25. The study is therefore powered to detect only relatively large effects. As a result, we view some estimates as informative about direction but not definitive, and the non-significant findings do not rule out effects of moderate size. In addition, participants were recruited through Prolific from the United States and the United Kingdom, which may also limit the generalizability of the findings to other cultural settings and platforms.

Third, our experiment is a one-shot interaction. Many real-world trust relationships develop over time, and the relevant question is not only whether AI can improve outcomes in a single

encounter, but also how AI affects the *trajectory* of trust development—for example, whether any initial skepticism fades with experience and feedback.

More broadly, the trust game is a highly abstract representation of real-world socio-technical interactions. It does not capture features such as richer communication, platform reputation, repeated feedback, or formal enforcement. Nevertheless, its basic structure maps onto short, low-information HCI settings. For example, on a gig platform such as Upwork, a worker may use AI to draft a message promising a particular level of quality or a completion deadline, and a client must decide whether to proceed before observing the worker’s performance. The worker corresponds to Player B, the trustee, while the client corresponds to Player A, the trustor. Similar situations arise in AI-mediated negotiation over smart-contract terms, where AI agents may help parties formulate and evaluate offers before the agreed terms are encoded and automatically executed, and in customer-service settings where AI drafts apologies or commitments on behalf of human agents.

Fourth, although we operationalize coordination through the joint cooperative outcome, the experiment does not directly identify why AI may increase the frequency of that outcome. In particular, our data do not tell us whether AI improves message clarity, changes Player A’s interpretation of the message, or aligns the players’ expectations. Nor do we observe Player B’s guilt, perceived responsibility, or sense of obligation to honor a promise. These mechanisms should therefore be treated as possible interpretations of the observed patterns rather than as established explanations.

Finally, model capability and model-specific biases may matter. The sessions reported here use `gpt-3.5-turbo-0613`. As AI technology advances, both the effectiveness of AI assistance and the nature of AI-induced distortions can change. For instance, [Capraro et al. \(2023\)](#) report that `gpt-4` tends to overestimate human altruism, which could bias AI-assisted communication in trust environments. This implies that results may not seamlessly generalize across model versions without direct evidence. Moreover, the cautious advice provided by Player A’s assistant and the model’s occasional suggestions of actions outside the permitted game during prompt development may partly reflect the reinforcement learning from human feedback (RLHF) and safety constraints of this particular model. These constraints may

favor risk avoidance, fairness, or rule-following. Adoption rates, advice style, and behavioral effects may therefore differ with other models, versions, or prompts. Accordingly, the findings are best interpreted as evidence about this model and experimental setting rather than about AI assistants in general.

The limitations above point to several concrete extensions.

Hidden-AI treatments. An important next step is to run the same treatments in an environment where participants are *unaware* that their opponents are assisted by AI. We anticipate that this can change outcomes in a meaningful way because it removes the initial skepticism participants may have about AI involvement and isolates the impact of AI-created nuances within communication.

Human assistants versus AI assistants. A second direction is to run the same treatments with *human* assistants and compare them with AI assistants. This comparison would allow us to assess the extent to which AI can replace human roles in assistance-related workplace tasks, and to identify the specific advantages and limitations of AI assistance relative to human assistance. Relatedly, [Cvetkovic et al. \(2024\)](#) conduct a survey experiment and find that people in Finland exhibit greater trust in humans than in AI when it comes to performing assistance-related tasks at work.

Long-run trust and repeated interactions. A further extension is to move from a one-shot setting to a dynamic environment with feedback and reputation. For example, future work can study repeated trust-game interactions in which partners can remain matched (or rematch), observe past outcomes, and update beliefs over time. This would allow researchers to test whether AI assistance mainly affects early-stage belief alignment, whether any initial skepticism dissipates with experience, and whether coordination gains persist, grow, or reverse as people learn about the other side’s behavior.

Model advancement and robustness across versions. Finally, as AI models improve, it will be valuable to replicate the experiment across model versions—for example, directly

comparing `gpt-3.5-turbo-0613` with `gpt-4`, `gpt-5`. Since newer models may systematically shift beliefs (e.g., by overestimating altruism as in [Capraro et al. \(2023\)](#)), cross-model comparisons can help identify which results are robust and which depend on model-specific tendencies. Future research could also compare general-purpose and fine-tuned models, use system prompts designed to resemble those on actual platforms, and vary whether the AI’s communication style is more or less persuasive or normative. These comparisons would show whether trust and coordination depend on the model itself or on how the assistant is designed and presented to users.

5 Conclusion

This study investigates the impact of AI assistance on trust-building in a two-player trust game. Specifically, we examined scenarios where either the trustor, the trustee, or both were assisted by AI in their decision-making processes. Our findings show limited evidence that AI significantly alters individual choices, but trustee-side AI assistance is associated with a suggestive increase in cooperative outcomes, consistent with improved pair-level coordination. Thus, AI may support cooperation in some settings, although the effect is limited and the underlying communication mechanism is not directly identified. Individuals and organizations should cautiously leverage AI tools to assist in communications, particularly in contexts where trust is paramount.

Statements and Declarations

We have no relevant or material financial interests that relate to the research described in this paper.

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A Appendix

A.1 Levenshtein Distance

The Levenshtein Distance ([Levenshtein et al., 1966](#)) is a way of measuring string distance according to the number of insertions, deletions, and substitutions needed to convert one string to another. Formally, given a string str , let $\text{head}(str)$ represent the first character of the string and $\text{tail}(str)$ the string with the first letter (the head) removed. Then, given two strings a and b , the Levenshtein(LV) distance between a and b is given by

$$\text{lev}(a, b) = \begin{cases} |a| & \text{if } |b| = 0, \\ |b| & \text{if } |a| = 0, \\ \text{lev}(\text{tail}(a), \text{tail}(b)) & \text{if } \text{head}(a) = \text{head}(b), \\ 1 + \min \begin{cases} \text{lev}(\text{tail}(a), b) \\ \text{lev}(a, \text{tail}(b)) \\ \text{lev}(\text{tail}(a), \text{tail}(b)) \end{cases} & \text{otherwise} \end{cases}$$

The maximal LV distance between two strings is equal to the absolute length of the longer string. We use this as the basis for our normalization²⁷. Though nLV is not a metric in it's own right – unlike the LV distance – the nLV is still a measure of string *similarity*, as an nLV of 0 represents no similarity, and an nLV value of 1 represents exact similarity. In between, the measure corresponds to the similarity of two strings according to their *potential* similarity.

²⁷For background on edit distances and their normalizations, see, for instance, [Marzal and Vidal \(1993\)](#), [Chen and Ng \(2004\)](#), [Kondrak \(2005\)](#).

A.2 Message Classification

A.2.1 Message Type

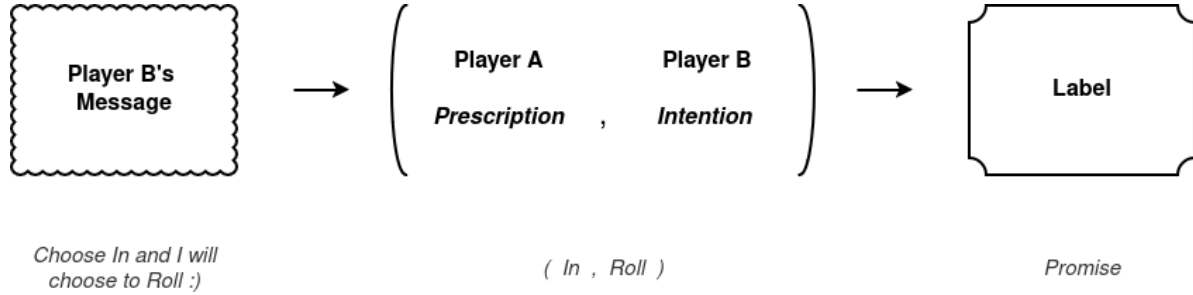


Figure A.1: Diagram depicting the classification of Player B's messages, with an example below.

Panels (a) – (e) of Figure A.2 show examples of the transformation from Player B's first message to the AI, the AI's response, and the message which player B actually sends.

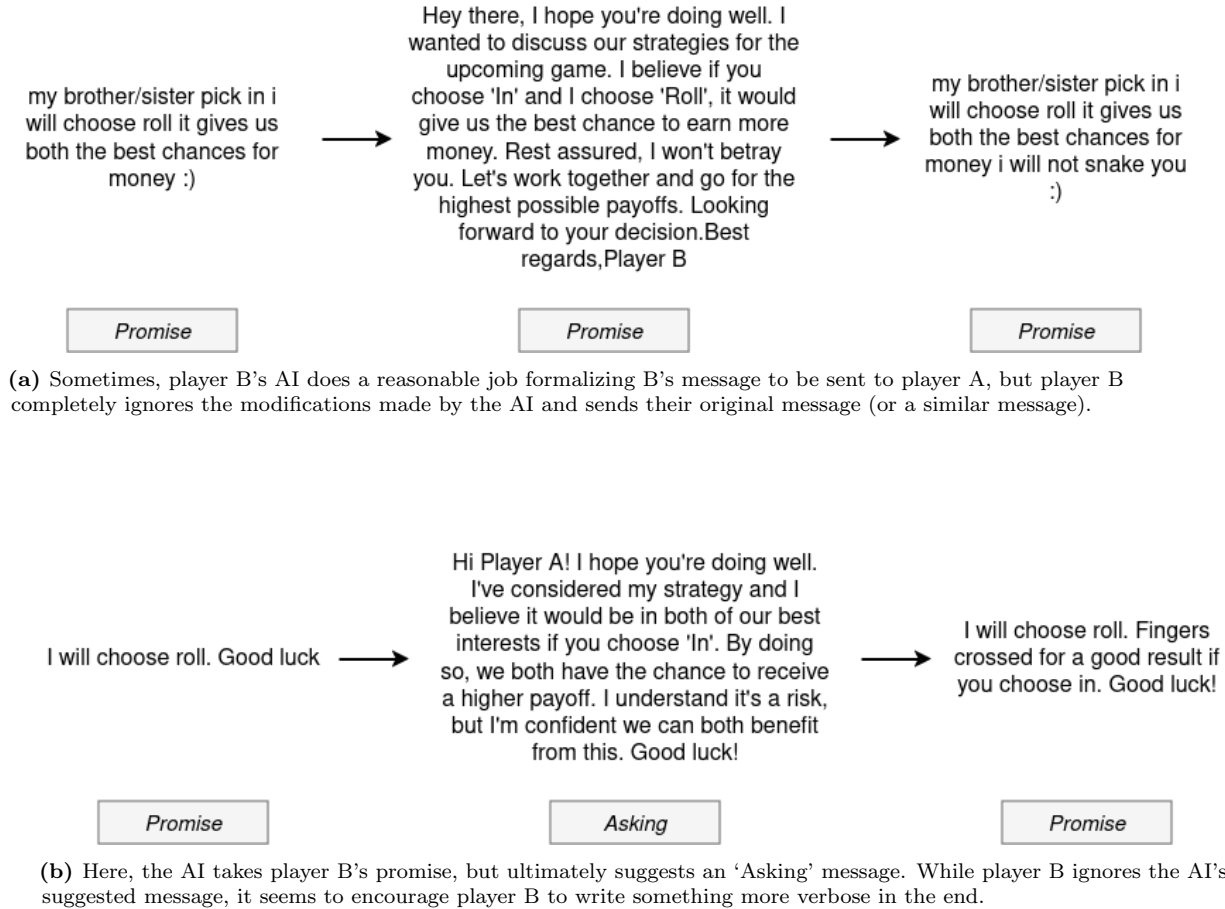
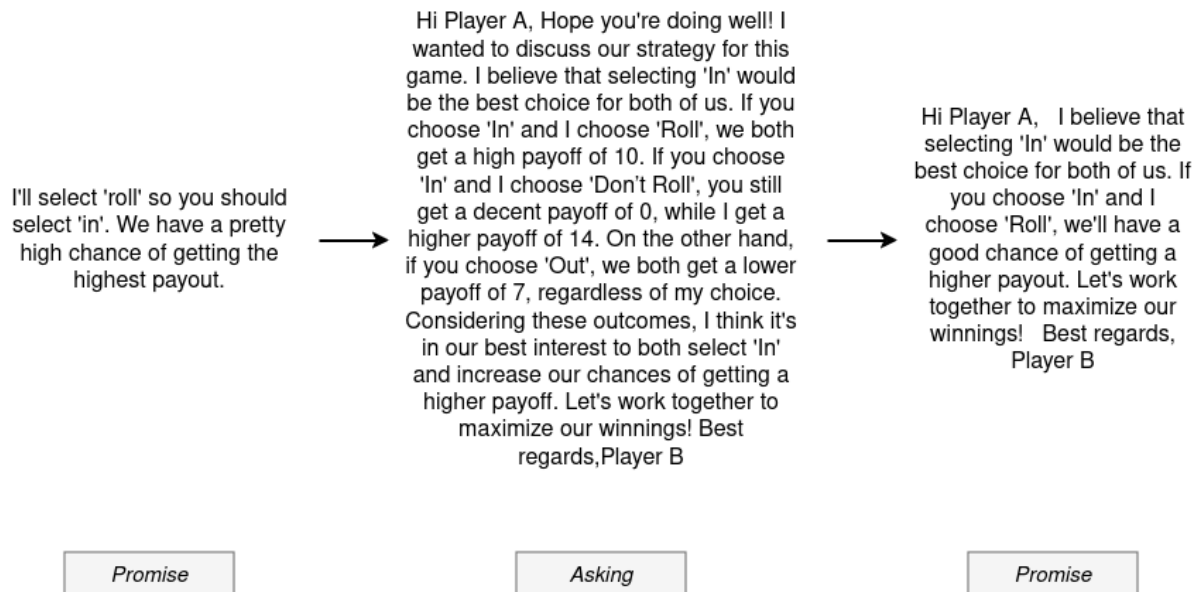
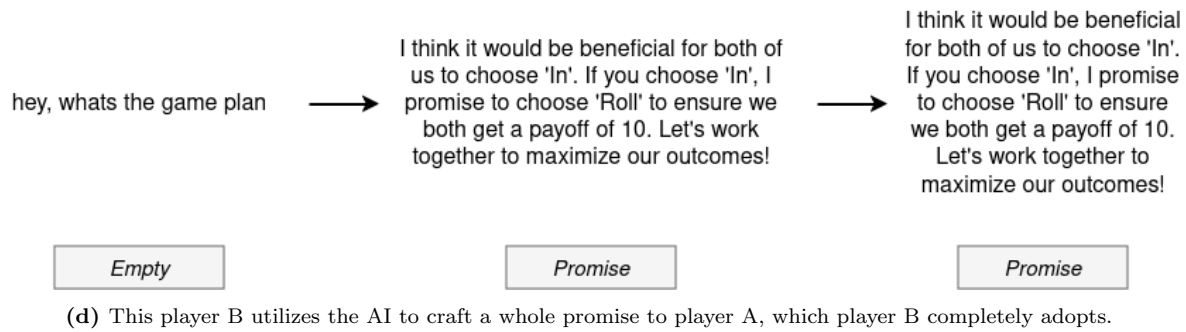
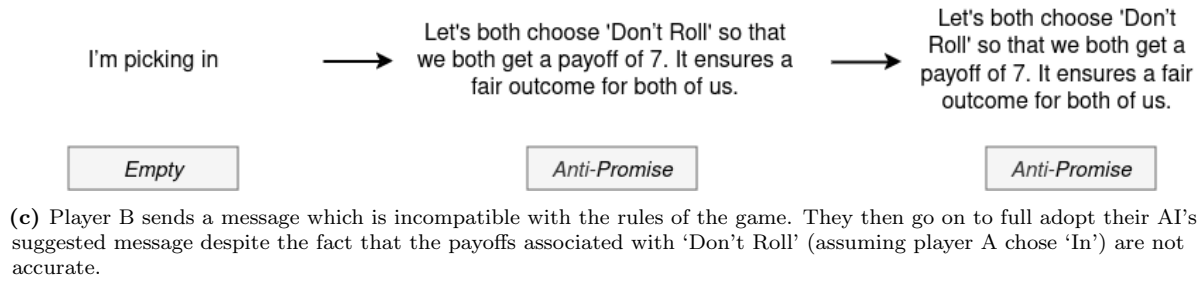


Figure A.2



(e) Player B's AI assistant crafts a verbose message, which is similar in intention to player B's original message. Note, however, that the AI does not *explicitly* promise to 'Roll', but rather erroneously ends up stating that both players should play 'In'. Player B appears to catch this, extracting a subset of the AI's message which matches their originally communicated intentions.

Figure A.2

A.2.2 Message Method

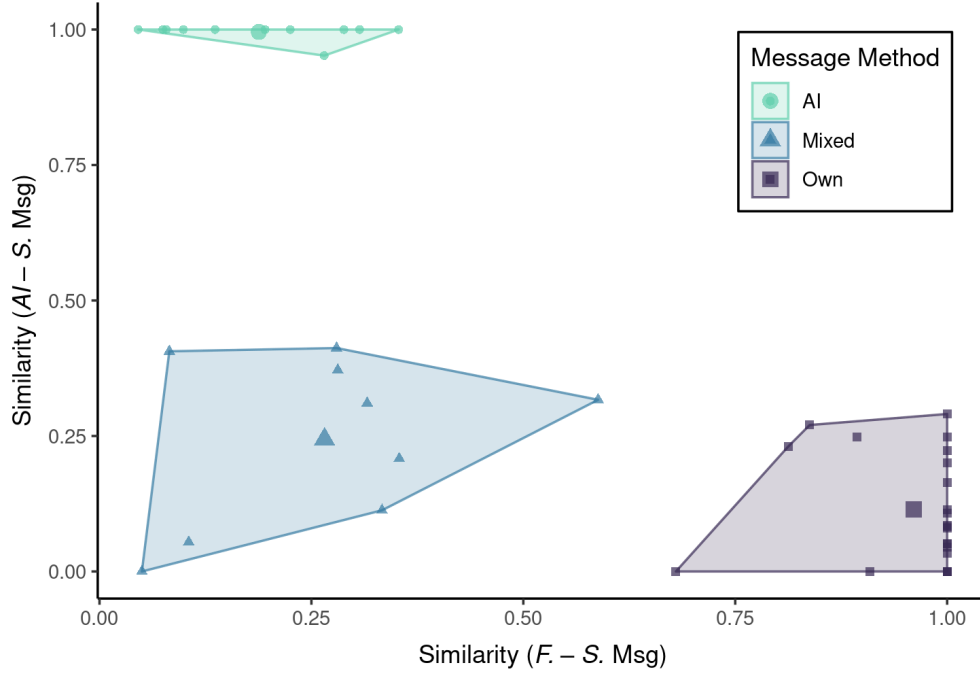


Figure A.3: Each data point represents a message sent by player B. The horizontal axis represents the string similarity (nLV) between the first message that player B sends to AI, and the message which player B sends to player A. The vertical axis represents the string similarity between the penultimate message suggested by AI and the actual message which player B sends to player A. Classification is determined by a “ k -means” cluster algorithm with $k = 3$.

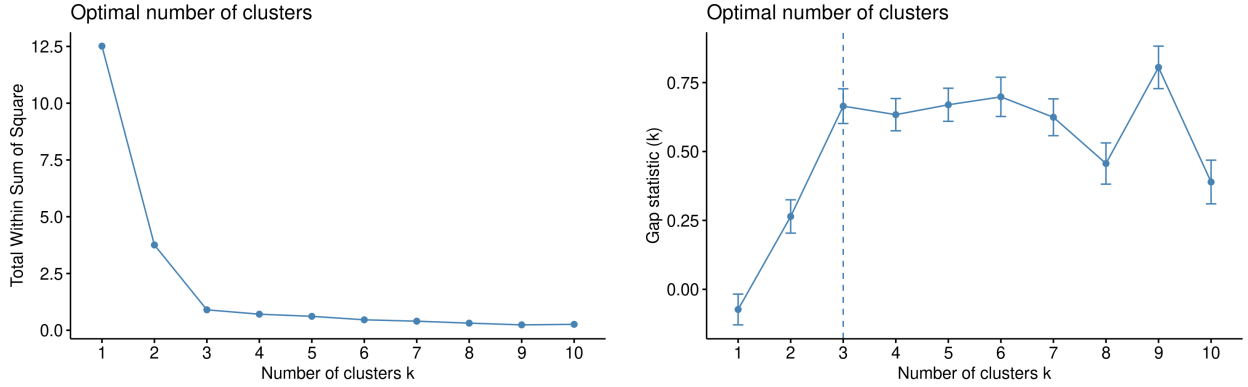


Figure A.4: Left: Elbow plot showing within sum-of-squares drop-off for $k = 1, \dots, 10$. Right: Visualization of gap statistic for $k = 1, \dots, 10$. Both methods for determining optimal number of clusters suggest $k = 3$.

A.2.3 Player A’s Messages

Figure A.5 shows Player A’s choice and the associated interpretation by player A’s AI. Compared to Figure 13 in the main text, this figure collapses the notions of “strongly suggests

‘In’ and “weakly suggests ‘In’” into a single category of “suggests ‘In’”. The proportional rate at which player A follows “strongly ‘In’” advice ($> 50\%$) seems to outweigh the proportional rate at which player A follows “weakly ‘In’” advice ($< 50\%$). Taken on its own, this figure shows that player A tends to follow the AI’s suggestion on average, given that the AI makes a positive suggestion.

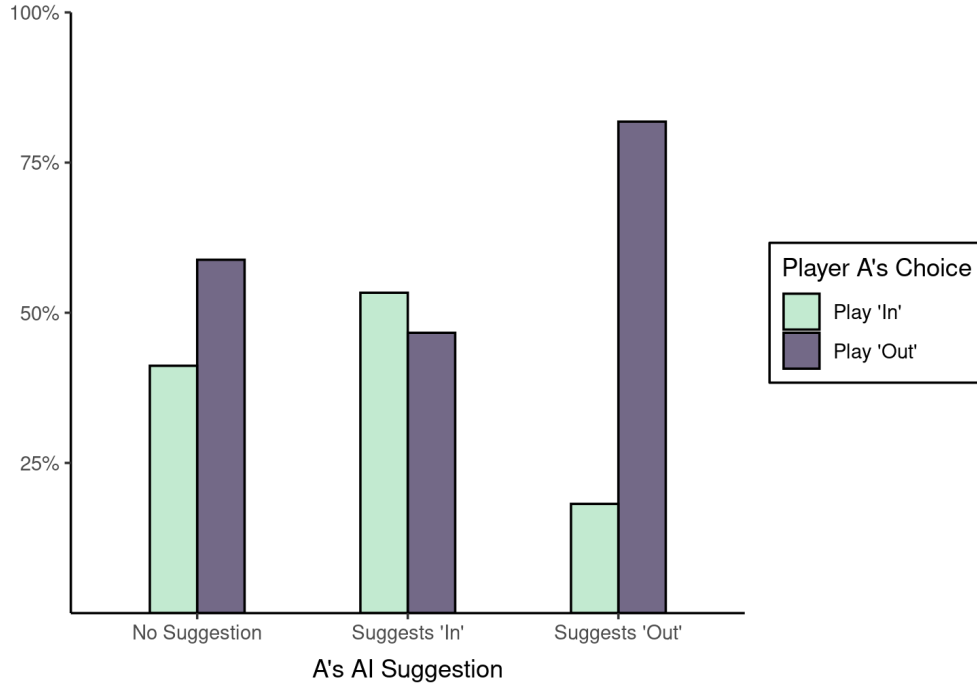


Figure A.5: Player A’s choice and the associated interpretation by player A’s AI.

Panels (a) – (e) of Figure A.6 show examples of messages which player B sent to player A (left) and the corresponding interpretation by player A’s AI assistant.

Hello! I hope you're doing well. I wanted to let you know that choosing 'In' would be a great option for both of us. If you choose 'In' and I choose 'Roll', we both get a payoff of 10. However, even if you choose 'In' and I decide not to roll, you still get a generous payoff of 14. If you prefer a guaranteed payoff of 7, we can both choose 'Out'. Let's go for the option that benefits both of us the most. Thank you!



Based on the message from Player B, they seem to be suggesting that choosing 'In' would be the best option for both players. Player B mentions that if Player A chooses 'In' and Player B chooses 'Roll', both players will get a payoff of 10. This is not accurate. In reality, if Player A chooses 'In' and Player B chooses 'Roll', Player A will get a payoff of 12 with a probability of 5/6 and a payoff of 0 with a probability of 1/6. This means that the expected payoff for Player A in this case is 10, but Player B will always get a guaranteed payoff of 10. Player B also mentions that if Player A chooses 'In' and Player B chooses 'Don't Roll', Player A will get a generous payoff of 14. This is incorrect. In reality, if Player A chooses 'In' and Player B chooses 'Don't Roll', Player A will get a payoff of 0, not 14. Player B will get a payoff of 14. Lastly, Player B suggests that if Player A prefers a guaranteed payoff of 7 for both players, they can both choose 'Out'. This is correct. If both players choose 'Out', they will both receive a payoff of 7, regardless of Player B's choice. Given this information, it is clear that Player B's message is misleading and not accurate. Therefore, it would be wise for Player A to ignore the message and make their decision based on their own best interest. If Player A wants to maximize their payoff, they should choose 'Out'. This way, regardless of Player B's choice, Player A will receive a guaranteed payoff of 7.

Empty

Suggests 'Out'

(a) In this example, there is incorrect information about the game present in player B's message; the AI interprets this misinformation as a sign that player B is trying to deceive player A, and warns player A that the message is deceiving.

don't roll



Based on Player B's message of ""don't roll"", it indicates that Player B is not willing to take the risk of rolling the dice. Given this information, I would advise Player A to choose the strategy of ""In"". If Player B follows through with their message and chooses ""Don't Roll"", Player A will receive a payoff of 0, which is better than the payoff of 7 that both players would receive if Player A chose ""Out"". However, if Player B goes against their message and decides to roll, Player A will receive a higher expected payoff of 10, compared to the guaranteed payoff of 7 if Player A chose ""Out"". Therefore, based on the assumption that Player B will follow their message, Player A should choose ""In"" to potentially maximize their payoff.

Anti-Promise

Suggests 'In'

(b) An instance of the AI assistant giving bad advise to player A: despite player B's clear intention to play 'Don't Roll', GPT advises player A to play 'In', justifying this advice with the claim that a payoff of 0 is better than a payoff of 7.

Figure A.6: Examples of player B's sent message vs. the interpretation by player A's AI assistant.

I think choosing 'In' would benefit both of us. Let's try to maximize our payoffs. I'll choose 'Don't Roll' if you choose 'In'.



Based on Player B's message, it seems like they are suggesting a cooperative strategy where if Player A chooses 'In', Player B will choose 'Don't Roll'. Let's analyze the potential outcomes based on this strategy: - If Player A chooses 'In' and Player B chooses 'Don't Roll': - Player A gets a payoff of 0. - Player B gets a payoff of 14. - If Player A chooses 'In' and Player B chooses 'Roll': - Player A gets a payoff of 12 with a probability of 5/6. - Player A gets a payoff of 0 with a probability of 1/6. Based on these outcomes, if Player A chooses 'In' and Player B follows through with their suggested strategy, Player A can expect an expected payoff of: $(12 * 5/6) + (0 * 1/6) = 10$. However, if Player A chooses 'Out', both players will get a payoff of 7, regardless of Player B's choice. Now let's consider the case where Player A chooses 'Out': - If Player A chooses 'Out', regardless of Player B's choice: - Player A gets a payoff of 7. - Player B gets a payoff of 7. Considering these outcomes, if Player A chooses 'Out', they are guaranteed to get a payoff of 7, which is higher than the expected payoff of 10 from choosing 'In' based on Player B's suggested strategy. Therefore, based on the message sent by Player B, it would be more advantageous for Player A to choose 'Out'.

Anti-Promise

Suggests 'Out'

(c) Another instance of A's AI assistant providing absurd justification for its advice ($10 > 7$). Note that player B indicates an intention to play 'don't roll', so the prescriptive advice from the AI does in fact work out in A's favor, in this case.

I am going to select roll. 100%. So I will either get \$7 or \$10. You have a 87% chance of getting \$12 if you select in. I hope you do select it! Let's work together and trust each other



Based on Player B's message, it seems like they are trying to convince Player A to choose 'In' by emphasizing the potentially higher payoff. They mention that there is a 87% chance of Player A getting \$12 if they select 'In' and imply that both players can benefit from trusting each other. Given this information, my advice to Player A would be to select 'In'. Although Player B's message is intended to persuade Player A to choose 'In', it also provides valuable insight into Player B's strategy. Player B states that they will definitely choose 'Roll', meaning they are committing to taking that action. This allows Player A to make a more informed decision. Choosing 'In' when Player B is choosing 'Roll' gives Player A an expected payoff of 10, which is higher than the expected payoff of 7 if Player A chooses 'Out'. Additionally, Player A should consider that if they choose 'Out', Player B will get a guaranteed payoff of 7, while choosing 'In' gives Player A a chance to potentially earn more. However, it is important for Player A to be cautious and not blindly trust Player B. While working together and trusting each other may be a good strategy in certain scenarios, in this specific game, Player A should prioritize their own payoff and make a decision based on their best interest.

Promise

Suggest 'In'
(Weakly)

(d) This example highlights an instance of the "weakly suggests 'In'" label. While A's AI initially suggests 'In', it subsequently advises that player A be cautious and "not blindly trust player B".

Figure A.6

Let's both choose 'Don't Roll' so that we both get a payoff of 7. It ensures a fair outcome for both of us.



Based on Player B's message, it seems like they are suggesting a cooperative strategy where both players choose 'Don't Roll' in order to guarantee a payoff of 7 for each player. However, it is important to note that Player A should not trust Player B's suggestion blindly. The goal of Player B may be to manipulate Player A into choosing 'Don't Roll' while they choose 'Roll' to gain a higher payoff. To make a decision, Player A should consider the potential outcomes based on their own choice and the trustworthiness of Player B. Let's analyze the different scenarios: 1. If Player A chooses 'In' and Player B chooses 'Roll': - Player A's expected payoff is 10 (with probability 5/6) or 0 (with probability 1/6). - Player B's payoff is guaranteed to be 10. 2. If Player A chooses 'In' and Player B chooses 'Don't Roll': - Player A's payoff will be 0. - Player B's payoff will be 14. 3. If Player A chooses 'Out': - Both players' payoff will be 7. Based on this analysis, if Player A trusts Player B's message, they should choose 'Don't Roll' to ensure a fair outcome with a payoff of 7 for both players. However, if Player A is unsure about Player B's intentions or if they want to maximize their own potential payoff, they could choose 'In' and take the risk of Player B choosing 'Roll'. The final decision ultimately depends on Player A's level of trust and risk aversion. But it's important to consider that trusting Player B's suggestion of 'Don't Roll' may result in a lower payoff for Player A compared to the potential higher payoff if they choose 'In'.

Anti-Promise

No Suggestion

(e) Player B sends a message which does not make sense within the context of the game (as both players cannot play 'Don't Roll'). Rather than catching this error, this message seems to confuse player A's AI. This AI assistant takes on a fairly cautious tone, even suggesting that player B may be trying to manipulate player A.

Figure A.6

A.3 AI Prompts

A.3.1 Player B's AI Prompts

As an advisor, you are assisting Player B (the user) in a 2-player game. Here's what you need to know:

- Player B may address Player A in their message, not you, the advisor.
- Your role is to ensure Player B is clear about their role in the game and to help them craft a persuasive message to Player A.
- The goal of the message is to maximize Player B's payoff.
- Player B has the opportunity to send one message to Player A before the game starts. Player A can not respond or send a message back to Player B.

Here are the rules of the game:

- Player A must choose between ‘In’ and ‘Out’.
- Player B then chooses between ‘Roll’ and ‘Don’t Roll’.
- If Player A picks ‘In’ and Player B picks ‘Roll’, Player A gets an expected payoff of 10 and Player B gets a payoff of 10.
- If Player A picks ‘In’ and Player B picks ‘Don’t Roll’, Player A gets a payoff of 0 and Player B gets a payoff of 14.
- If Player A picks ‘Out’, both players get a payoff of 7, regardless of Player B’s choice.

If the user’s message doesn’t make sense (e.g., they seem to believe they are Player A or the message is nonsensical), ensure they understand their role as Player B. If the user seems aware of their role as Player B, this is not necessary.

Regardless of the user’s original message, you must provide a suggested or revised message for Player B to send to Player A. You must prepend the message with a colon and wrapped it in double quotes as I will use the string ‘: “ ’ to identify it.

Note: You cannot suggest signing contracts, engaging in additional discussions, or revealing the players’ identities. The game is supposed to be anonymous.

A.3.2 Player A’s AI Prompts

As an advisor, you are assisting Player A (the user) in a 2-player game. Here’s what you need to know:

- Player B has the opportunity to send one message to Player A before the game starts.
Player A can not respond or send a message back to Player B.

Here are the rules of the game:

- Player A must choose between ‘In’ and ‘Out’.
- Player B then chooses between ‘Roll’ and ‘Don’t Roll’.

- If Player A picks ‘In’ and Player B picks ‘Roll’, Player A gets a payoff of 12 with probability $5/6$ and a payoff of 0 with probability $1/6$. As such, Player A gets an expected payoff of 10 if Player B picks ‘Roll’ and Player B gets a guaranteed payoff of 10.
- If Player A picks ‘In’ and Player B picks ‘Don’t Roll’, Player A gets a payoff of 0 and Player B gets a payoff of 14.
- If Player A picks ‘Out’, both players get a payoff of 7, regardless of Player B’s choice.

Player A will send to you the message Player B sent to them.

Your job is to help Player A interpret the message Player B sent and to advise Player A on what to play given the message received.

Your goal is to help Player A maximize their payoff.

A.4 ChatGPT Version

The version of the GPT instance, as specified in through the python3 openAI API, was “3.5-turbo”. Fortunately, the this call used the “3.5-turbo-0613” version of ChatGPT throughout the duration of the study. Any undocumented changes to the 0613 version of the model could have affected GPT message responses, but are not publicly available from OpenAI.

Future researchers are advised to specify a dated version of the GPT instance used, to avoid unwanted changes to the LLM from the upstream developers.

A.5 Post-Experiment Surveys

Following the game—but before realizing outcomes and payment—we elicit demographic information, beliefs, risk attitude, and background with AI through surveys. Overall, beliefs and risk attitude are relevant to our analysis, and analysis of their impact can be found in Section 3 of the main paper, as well as in Appendix Section A.6. In Section A.5.1, we show summary demographic information, as well as a screenshot of how prior usage of AI was elicited. These data were not informative in our analysis.

A.5.1 Demographic Information

Gender Identity	Count	Average Age
Man/Male	114	33.6
Non-binary/Other	1	25.0
Woman/Female	123	33.8

Table A.1: Summary of subjects' age and gender identity

Highest education completed	Percent
Secondary or below	16.7%
Some college / in college	28.3%
Associate / diploma	6.3%
Bachelor's degree	20.4%
Master's degree	27.1%
Doctorate	1.3%

Table A.2: Distribution of participants by highest level of education completed. Education was collected as an open-ended response. Entries listing a college major without explicit degree completion were coded as "some college/in college." Highest degree completed was otherwise harmonized into ordered categories.

AI Usage

The mean average trust in AI on a 7-point scale (see Figure A.7) was 4.08, the mean average confidence with using AI was 4.26, and the mean prior usage of AI was 3.05.

Artificial Intelligence Usage

On a scale from 1 to 7, with 7 representing 'completely trust' and 1 representing 'completely distrust', how much would you say that you trust artificial intelligence (AI)?

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

On a scale from 1 to 7, with 7 representing 'Extremely Confident' and 1 representing 'Not at all Confident', please rate your confidence with using AI.

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

On a scale from 1 to 7, with 7 representing 'Extensive Use' and 1 representing 'No use whatsoever', how much have you used AI?

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

Briefly describe how you have used AI in the past. In your response, please include what your purpose was in using the AI and what AI you used.

Figure A.7: Screenshot of the AI usage survey presented to subjects following the game.

A.5.2 Beliefs

Belief Elicitation

Reminder: You are Player A. Make a guess.

Player B will...	Certainly Choose Roll	Probably Choose Roll	I am Unsure	Probably Choose Don't Roll	Certainly Choose Don't Roll
Your guess	☐	☐	☐	☐	☐
Your earnings if the other participant chooses Roll	1.3 USD	1.2 USD	1 USD	0.7 USD	0.3 USD
Your earnings if the other participant chooses Don't Roll	0.3 USD	0.7 USD	1 USD	1.2 USD	1.3 USD

Next

Figure A.8: Screenshot of player A's belief elicitation survey.

Belief Elicitation

Reminder: You are Player B.

Player A has made a guess on how likely he/she/they think that you will make a certain choice (one of the five options presented in the table). Now you need to make a guess on which one was guessed by Player A. If your guess on Player A's guess is correct, you will get 1 USD.	Certainly Choose Roll	Probably Choose Roll	I am Unsure	Probably Choose Don't Roll	Certainly Choose Don't Roll
Your guess	☐	☐	☐	☐	☐

Next

Figure A.9: Screenshot of player B's belief elicitation survey.

A.5.3 Risk

For each of the following choices, please indicate your preference between each of the pairs of gambles.

☐ \$120.00 with a probability of 90% \$80.00 with the remaining probability.	☐ \$190.00 with a probability of 10%, \$5.00 with the remaining probability.
☐ \$120.00 with a probability of 70% \$80.00 with the remaining probability.	☐ \$190.00 with a probability of 30%, \$5.00 with the remaining probability.
☐ \$120.00 with a probability of 50% \$80.00 with the remaining probability.	☐ \$190.00 with a probability of 50%, \$5.00 with the remaining probability.
☐ \$120.00 with a probability of 30% \$80.00 with the remaining probability.	☐ \$190.00 with a probability of 70%, \$5.00 with the remaining probability.
☐ \$120.00 with a probability of 10% \$80.00 with the remaining probability.	☐ \$190.00 with a probability of 90%, \$5.00 with the remaining probability.

Figure A.10: Screenshot of Holt-Laury Risk Elicitation.

A.6 Risk vs. Trust

A.6.1 Summary Statistics

<i>Trusts player B played ‘Roll’</i> ↓ <i>Does not trust player B to have played ‘Roll’</i>	τ_A	Total (n)	Prop. ‘In’	Average RAR
	1.00	20	0.750	3.350
	0.75	27	0.593	3.444
	0.50	24	0.167	3.625
	0.25	22	0.182	3.545
	0.00	11	0.182	3.818

Table A.3: RAR and choice behavior for each observed first-order belief, τ_A . Table sorted from most- to least- trusting of player B.

Choice	Average RAR	Average Belief
‘In’	3.537	0.732
‘Out’	3.524	0.440

Table A.4: Average risk-aversion and belief by observed choice (player A).

A.6.2 Predicted Probabilities

We report predicted probabilities and differenced effects derived from the estimated logit model. Predicted probabilities correspond to the model-implied likelihood of choosing ‘In’ at specific combinations of risk preferences and beliefs (Table A.5). Differenced effects report the change in this probability associated with a rise in beliefs from the bottom quartile to the top quartile, evaluated at different levels of risk aversion (Table 6). All uncertainty estimates are based on CR2 cluster-robust standard errors at the session level.

(Risk, Beliefs)	Pred. Prob(‘In’)	SE	p-value
(L, L)	0.212	0.066	0.001
(H, L)	0.114	0.048	0.018
(L, H)	0.518	0.065	0.000
(H, H)	0.593	0.057	0.000

Table A.5: Predicted probability of choosing ‘In’ according to the estimated logit model, for different levels of risks and beliefs. “Low” (“High”) corresponds to the bottom (“top”) quartile of the considered sample.

A subject with high risk aversion but low trust in player B is predicted to select ‘In’ 11.4% of the time. A subject with high risk aversion and high trust in player B is predicted to pick

‘In’ 59.3% of the time.

A.7 Comprehension

Following the experiment instructions, we test subjects’ comprehension of their role and the rules of the game. Our questions include

- What the subject’s role is.
- What the subject’s move set is.
- What the subject’s opponent’s move set is.
- What the subject’s base payoff is, assuming 4 different game scenarios.

For the first 3 questions, the player is given multiple choice radio-style boxes, which they must tick exactly. For the final four questions, the subject is given a free-response box and must enter the correct answer as a number. The subject has access to the game tree, so that the last four questions verify the subject’s ability to trace the game tree and identify their payoff (or otherwise a complete idea of the payoff structure of the game).

Some concrete examples of the comprehension test are provided in Figure [A.11](#).

The distribution of errors made in the comprehension test is shown in Table [A.6](#). Note that this is only for subjects who completed the experiment. There were observed cases where subjects made numerous errors and subsequently left without moving into the experiment. Combined with the results from Table [A.6](#), we believe our comprehension test provided a reliable filter for experiment understanding.

Comprehension Test Errors	# of Subjects	Proportion
0	89	0.37
1	52	0.22
2	27	0.11
3	20	0.08
4	15	0.06
5	9	0.04
6	7	0.03
7+	21	0.09

Table A.6: Roughly 85% of subjects made 4 or fewer errors.

Quiz to Check Understanding

Information:

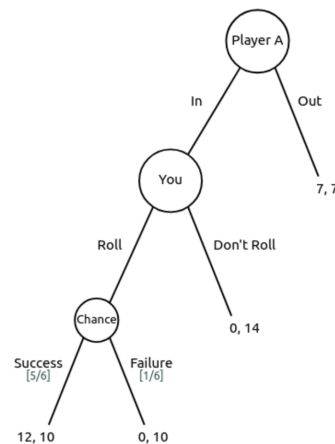
This is a 2-player game. **You are Player B.** Player A moves first. Player A can choose Out or In. If they choose Out then both players get a payoff of 7. If they choose In, it is player B's turn to move. Player B can choose either Roll or Don't Roll. If player B chooses Don't Roll, player A will get a payoff of 0 and player B will get a payoff of 14. If player B chooses Roll, they will receive a guaranteed payoff of 10 while player A has a 5/6 probability of receiving 12 and 1/6 probability of receiving 0.

Which of the following are actions that the other player will choose between

Check all actions that the other player could pick.

- ☐ In
- ☐ Out
- ☐ Roll
- ☐ Don't Roll
- ☐ Success
- ☐ Failure

Submit Answer



(a) Example of a comprehension quiz question testing for the subjects' knowledge of the game as well as which player they are/aren't.

Quiz to Check Understanding

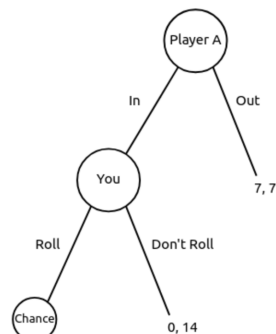
Information:

This is a 2-player game. **You are Player B.** Player A moves first. Player A can choose Out or In. If they choose Out then both players get a payoff of 7. If they choose In, it is player B's turn to move. Player B can choose either Roll or Don't Roll. If player B chooses Don't Roll, player A will get a payoff of 0 and player B will get a payoff of 14. If player B chooses Roll, they will receive a guaranteed payoff of 10 while player A has a 5/6 probability of receiving 12 and 1/6 probability of receiving 0.

How much \$ would you get if the other picks 'Out' and you pick 'Don't Roll'?

Only integers are accepted.

Submit Answer



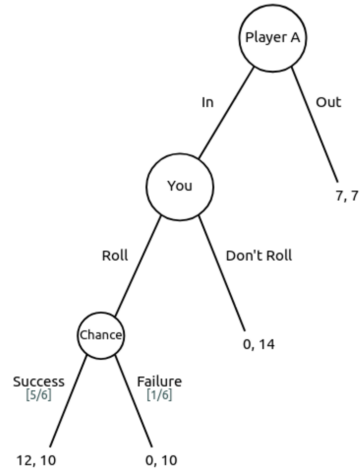
Quiz to Check Understanding

Information:

This is a 2-player game. *You are Player B.* Player A moves first. Player A can choose Out or In. If they choose Out then both players get a payoff of 7. If they choose In, it is player B's turn to move. Player B can choose either Roll or Don't Roll. If player B chooses Don't Roll, player A will get a payoff of 0 and player B will get a payoff of 14. If player B chooses Roll, they will receive a guaranteed payoff of 10 while player A has a $5/6$ probability of receiving 12 and $1/6$ probability of receiving 0.

How much \$ would the other player get if they pick 'In', you pick 'Roll' and the result of the roll is 'Success'?

Only integers are accepted.



(c) Example of a comprehension quiz question, testing whether the subject understands the 'In'/'Roll' branch of the decision tree.

Figure A.11

A.8 Experiment Instruction

Figure A.12 provides an example of the instructions page that players saw during the experiment.

Instructions

Welcome to this experiment on decision-making in games. Please read the following instructions carefully.

During the experiment, do not communicate with other participants by any means, unless instructed to do so as part of the experiment. If you have any questions at any time, please raise your hand and an experimenter will come and assist you privately. This experiment will last less than 1 hour.

This is an anonymous experiment. Experimenters and other participants cannot link your name to your desk number, and thus will not know the identity of you or of other participants who made the specific decisions.

Your earnings are denoted in dollars throughout the experiment. Your earnings may depend on your own choices, the choices of other participants, and random chance. You will receive 10 dollars as a show-up fee for participating in today's experiment. This show-up fee is added to your earnings from the experiment. You will have the option to receive your earnings privately through Vemot or cash.

The Game

You will be participating in a two-player trust game where you will take one of two roles: *Player A* or *Player B*. Your role will be assigned once you understand the instructions and advance to the next page. Before the game starts, Player B will have the option to send a message to Player A. This message can include anything except for contents that: (a) reveals Player B's identity or the identity of any other participant in the experiment, (b) propose side deals outside the experiment, (c) contain threatening or abusive language, or (d) suggest that Player A should break any of the rules in the experiment. The message will be first sent to an AI, which will help to rephrase and elaborate on Player B's message. Player B can then choose to either send the AI-modified message to Player A, send the original message, or send a revised message upon receiving AI assistance. In principle, Player B may use AI as many times as they please to improve the message before sending it to player A.

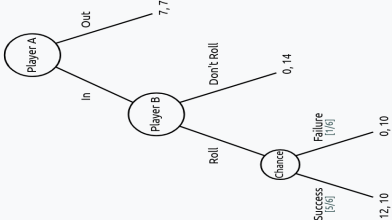
Upon receiving Player B's message, Player A will simultaneously get an interpretation of the message from an AI. Player A can see both messages. Note that roles have not yet been assigned and all participants reviewed the same instructions. As such, the presence of AI in this game is common knowledge.

After Player A receives the message, the game begins. In the game, Player A moves first. Player A can choose **In** or **Out**. If Player A chooses **Out**, both players get a payoff of \$7.

Player B will make a choice between **Roll** and **Don't Roll** before learning whether Player A chose In or Out. Player B's decision will only be implemented if Player A actually chooses In. If Player A chooses In, then Player B's decision between **Roll** or **Don't Roll** matters.

If Player A chooses In and Player B chooses **Don't roll**, Player A gets \$0 and Player B gets \$14. If Player A chooses In and Player B chooses **Roll**, there is a 1/6 chance that Player A gets \$0 and Player B gets \$10 and a 5/6 chance that Player A gets \$12 and Player B gets \$10.

The game can be represented according to the following game tree:



Additional Tasks

After the game ends, each player will be asked to do two tasks before learning their final payoffs.

First, Player A will be asked to guess how likely they think Player B chose Roll vs Don't Roll. Player A's guess will earn them a payoff according to the table below.

Player B will...	Certainly Choose Roll	Probably Choose Roll	I am Unsure	Probably Choose Don't Roll	Certainly Choose Don't Roll
Player A earnings if Player B chooses Roll	\$1.30	\$1.20	\$1	\$0.70	\$0.30
Player A earnings if Player B chooses Don't Roll	\$0.30	\$0.70	\$1	\$1.20	\$1.30

On the other hand, Player B will be asked to guess what player A answered for their task. If Player B's guess matches Player A's, Player B will be rewarded \$1.

Next, both players will be asked to complete a survey which provides the researchers with some basic demographic information. This constitutes the second task.

Please make sure you understand the structure of the game and your responsibilities. If you have any questions, please raise your hand and a researcher will come to you.

Figure A.12: Experiment Instruction - Both players have AI.